The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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Abstract

Objective. Temporal-interference (TI) stimulation applies two high-frequency currents with a low difference frequency. The field contains no component at that difference frequency. We examine how effective cellular nonlinearities generate this component under weak stimulation and how recurrent circuits amplify it. **Approach.** We drive a Jansen–Rit column, an exact quadratic integrate-and-fire (QIF) mean field, and its spiking network with two-tone inputs. We separate carrier access, nonlinear mixing, and recurrent gain. A weak-input expansion combines cellular contributions into an effective second-order response, realized by both models. **Main results.** For a smooth response with a nonzero second-order term, the leading beat scales with the product of the effective carrier amplitudes. A static rate curve mixes through its curvature; the exact mean field recovers this description when its membrane variables relax faster than the input, leaving dynamic synapses. Outside that limit the mixing coefficient depends on the population dynamics, and the membrane's carrier attenuation enters only once, through access. Near a Hopf bifurcation, the recurrent network selectively amplifies the generated beat as $1/\gamma$ within linear response, where γ is the modal damping. Coupling sweeps separate this gain from leading-order beat generation. Beat-locked rate modulation grows faster than the mean-rate change in the mean field and in a network of 8000 neurons per population. **Significance.** Fast cellular access and slow circuit resonance can coexist in one population model. The weak-input product law makes an envelope-only dose map conditional on the transduction mechanism. TI and direct low-frequency stimulation share the amplifier when effective input coupling, readout, and state are matched. The analysis concerns weak modulation of ongoing activity; biological access and absolute response at human field strengths remain unresolved.

Keywords: temporal interference; non-invasive neuromodulation; transcranial stimulation; neural mass model; amplitude demodulation; Hopf bifurcation; cross-frequency coupling; criticality.

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1 Introduction

Temporal-interference (TI) stimulation was proposed to reach deep brain targets while reducing stimulation of surrounding tissue [1]. It applies two sinusoids at f_1 and $f_2 = f_1 + \Delta f$ through separate electrode pairs, typically with $f_1 \sim 1$–2 kHz and $\Delta f \sim 1$–100 Hz. Where the fields overlap, their superposition is an amplitude-modulated (AM) signal with an envelope at Δf. Electrode currents can steer the region of largest envelope modulation. Whether this also localizes the neural response depends on how neurons transduce the waveform.

The difference frequency is absent from the input itself:

$$s(t) = \varepsilon_1 \cos(\omega_1 t) + \varepsilon_2 \cos(\omega_2 t), \qquad \omega_{1,2} = 2\pi f_{1,2}, \quad \Omega = \omega_2 - \omega_1 = 2\pi\,\Delta f. \tag{1}$$

Here $\varepsilon_{1,2}$ are the effective input amplitudes (related to the applied field in §2.2), Ω is the beat angular frequency, and $f_c = (f_1 + f_2)/2$ is the carrier frequency. The spectrum contains only the two applied lines (Fig. 3). A linear, time-invariant system can attenuate or phase-shift them but cannot create a new one. Passive membrane filtering therefore cannot demodulate TI [2]; a nonlinear element must mix the carriers into the low-frequency band, as in AM reception [3].

We consider weak modulation of ongoing neural activity. For a smooth response about an operating state, a nonzero second-order term mixes the carriers into their difference frequency. Cellular contributions can therefore be combined in an effective population response without identifying each microscopic mechanism. Direct stimulation thresholds and conduction block under strong TI lie outside this scope.

Voltage-gated ion channels provide a cellular mixing mechanism in Hodgkin–Huxley and McIntyre–Richardson–Grill (MRG) myelinated-axon models [2]. Luff et al. [4] directly observed subthreshold difference-frequency oscillations in individual neurons, including carriers near 5 kHz. Mixing persisted during synaptic blockade but was suppressed by blocking tetrodotoxin-sensitive sodium channels. Other preparations show carrier-driven excitation or block rather than envelope extraction [5, 6, 7]. Primate recordings at human-comparable fields report changes in spike timing without a corresponding mean-rate change, with weaker effects than conventional low-frequency stimulation [8]. In slices, pyramidal and parvalbumin (PV) neurons respond differently, and most pyramidal TI responses disappear under synaptic blockade [9]. Computational studies also find network-mediated responses [10, 11, 12]. These findings distinguish cellular mixing from the network response it can recruit.

We separate carrier access, nonlinear beat generation, and recurrent amplification (Fig. 1). The first calculation uses a Jansen–Rit firing-rate sigmoid: its curvature mixes the carriers, and a network with second-order synapses amplifies the beat near a Hopf bifurcation. An exact quadratic integrate-and-fire (QIF) mean field connects this static description to cellular dynamics. When membrane relaxation is fast relative to the input, a static population transfer curve emerges while the synapses remain dynamic; otherwise the mixing coefficient is frequency dependent. Simulations in the corresponding spiking network test the population description, and a laminar model extends the calculation to two resonant bands at low carriers.

Related neural-mass accounts use the same nonlinear mixing for endogenous cross-frequency coupling and nested amplitude modulation [13, 14]. Here the input is external, making the field-to-cell coupling an additional requirement. The population reduction identifies what follows once the carrier reaches the nonlinearity; it does not identify the anatomical access pathway.

2 Theory

2.1 Population transfer and the slow-synapse limit

A neural mass model describes a population through averaged variables [15, 16]. In the Jansen–Rit (JR) family, a linear synaptic filter converts presynaptic firing rate $r(t)$ into postsynaptic potential

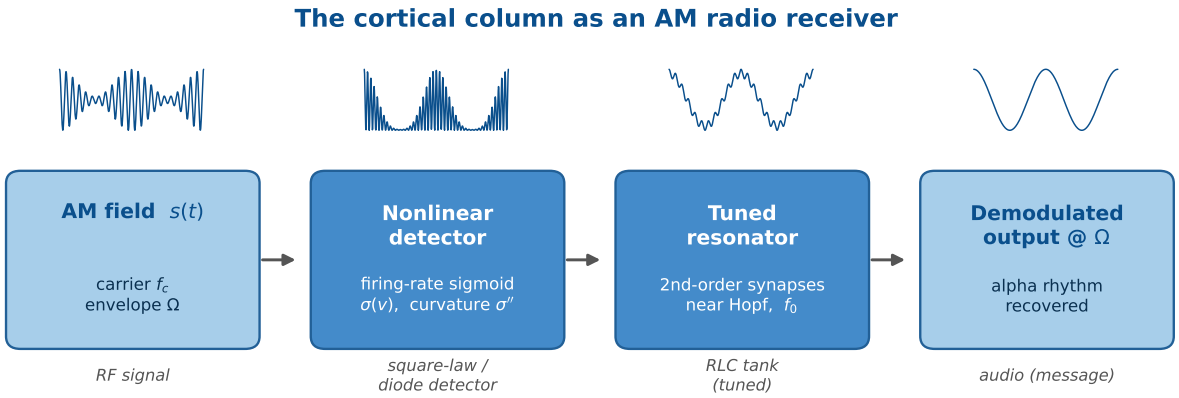


Figure 1: The cortical column as a tuned receiver. A nonlinear detector generates the beat at angular frequency Ω from a carrier near f_c; a recurrent network amplifies it near its natural frequency f_0. The diagram shows the static-sigmoid realization. In a conventional AM receiver, the tuned circuit selects the carrier before detection [3]; here the resonator selects the low-frequency output. The laminar extension also has a carrier-selective stage (§E).

$x(t)$:

$$\ddot{x} + 2a\dot{x} + a^2 x = Aa\, r(t),$$

with inverse synaptic time a and gain parameter A; inhibitory synapses use parameters b and B. Summed postsynaptic potentials give the mean input v, which a static transfer curve converts to firing rate [17],

$$\sigma(v) = \frac{2e_0}{1 + e^{\,\rho(v_0 - v)}}. \tag{2}$$

Here $2e_0$ is the maximum rate, v_0 the inflection voltage, and ρ the inverse voltage scale. Nonzero curvature makes this curve a mixer, while its slope sets the local gain of the recurrent circuit.

For a synapse from population Y to X, the resulting NMM1 equations are [15]

$$\hat{L}_{X\leftarrow Y}[u_{X\leftarrow Y}] = C_{X\leftarrow Y}\, r_Y, \qquad v_X = \Lambda[\hat{E}] + \sum_{Y:\, C_{X\leftarrow Y}\neq 0} u_{X\leftarrow Y}, \qquad r_X = \sigma(v_X). \tag{3}$$

The operator $\hat{L}$ is the inverse synaptic filter, $u_{X\leftarrow Y}$ its postsynaptic potential, $C_{X\leftarrow Y}$ the connectivity, and $\Lambda[\hat{E}]$ the direct polarization caused by the applied field. This field contribution enters alongside the synaptic potentials, without first passing through a synapse.

The corresponding population nonlinearity can be derived for all-to-all coupled QIF neurons with Lorentzian-distributed excitabilities. Their exact Montbrió–Pazó–Roxin (MPR) mean field evolves the firing rate r and mean voltage v [18]; adding dynamic synapses gives NMM2 [19, 20]. In membrane-time units,

$$\dot{r} = \frac{\Delta}{\pi} + 2rv, \qquad \dot{v} = v^2 + \bar{\eta} - \pi^2 r^2 + I(t), \tag{4}$$

where $\bar{\eta}$ is the mean excitability, $\Delta > 0$ its half-width, and $I(t)$ the normalized input, including recurrent drive when present. Rate and voltage are expressed in the corresponding rescaled units. For a constant total drive $Q = \bar{\eta} + I$, setting both derivatives to zero gives $v = -\Delta/(2\pi r)$ and

$$r = F(Q) = \frac{1}{\pi}\sqrt{\frac{Q + \sqrt{Q^2 + \Delta^2}}{2}}. \tag{5}$$

Equivalently, F averages the single-neuron steady rate $f(q) = \sqrt{q}/\pi$ for $q > 0$, and $f(q) = 0$ otherwise, over the excitability distribution. Here η is the neuron's excitability and $q = \eta + I$

its constant total drive. Threshold recruitment and the concave rate increase of active cells both enter its curvature. This nonlinear curve exists without recurrent coupling. It has an inflection at $Q = \Delta/\sqrt{3}$ and an unbounded concave rise, unlike the saturating JR sigmoid.

If the population relaxes to this equilibrium faster than synaptic inputs change, NMM2 reduces to a static rate map with dynamic synapses [19, 20]. Restoring the membrane time τ_m and physical firing rate, a second-order synapse with time constant τ_s obeys

$$r(t) \simeq \frac{F(Q(t))}{\tau_m}, \qquad \tau_s^2 \ddot{s}_{\mathrm{syn}} + 2\tau_s \dot{s}_{\mathrm{syn}} + s_{\mathrm{syn}} = r(t), \tag{6}$$

where s_{syn} is the filtered rate entering the recurrent drive in Q, with the original coupling normalization retained. This is the slow-synapse limit: shortening the effective membrane time makes the transfer appear static while the network retains its synaptic dynamics. The relevant separation is between the population relaxation time τ_{rel} and τ_s. Applying the same static map directly to a carrier of angular frequency ω_c requires the stronger condition $\omega_c \tau_{\mathrm{rel}} \ll 1$. A population can therefore be quasistatic for its synaptic inputs yet dynamic for a kHz carrier.

The transfer curve summarizes cellular nonlinearities within this regime. Its curvature combines their leading contributions to mixing; its slope converts a slow input into a rate change and enters recurrent gain. QIF supplies a derived realization through its spike-generating quadratic and excitability distribution. The generic weak-input mixing law does not require identifying each channel's contribution, although the coefficient and its frequency dependence are model-specific. Outside the static limit these contributions belong to a dynamic input–output functional. Section 2.3 calculates this response for QIF and recovers Eq. (5) in the adiabatic limit.

Cellular and circuit resonance are also distinct. Conductances such as I_h and I_M can tune single neurons [21]; the companion manuscript of de Palma Aristides et al. [22] reports cell-type- and state-dependent TI frequency selectivity in hippocampal neurons, sensitive to these conductances. Recurrent excitatory–inhibitory circuits can generate resonance without intrinsic cellular resonance [23, 24]. In the fast-membrane limit, any retained resonance belongs to the remaining synaptic circuit; modes requiring finite membrane dynamics need not persist.

The JR column couples pyramidal, excitatory-interneuron, and inhibitory-interneuron populations, with field input to the pyramidal sigmoid and local field potential (LFP) proxy $v = y_1 - y_2$ (App. B). The laminar extension (LaNMM) [13, 25] adds coupled alpha and gamma circuits: deep pyramidal P_1, spiny-stellate (SS), and somatostatin (SST) populations, and a superficial pyramidal P_2/PV circuit. These models retain dynamic synapses and static rate curves; NMM2 and its spiking realization retain membrane dynamics as well.

2.2 From applied field to a squared-envelope beat component

We use the quasistatic electromagnetic approximation for the applied field $E(t)$ (V/m), neglecting propagation and induction. This does not make membrane charging instantaneous. Let $s(t)$ be the polarization after the cellular access response; the static population model reads

$$r(t) = \sigma\big(v(t) + s(t)\big), \qquad E(t) = E_1 \cos \omega_1 t + E_2 \cos \omega_2 t. \tag{7}$$

The field acts directly on soma, dendrites, axon initial segment (AIS), and terminals, each with its own coupling and frequency response. In a reduced description, their contributions to the effective input are represented by parallel entry points,

$$s(t) = \sum_c \boldsymbol{\lambda}_f^c \cdot \mathbf{E}(t), \qquad \boldsymbol{\lambda}_f^c = \boldsymbol{\lambda}_0^c \, |H_c(\omega_c)|, \tag{8}$$

where $\boldsymbol{\lambda}_0^c$ is the static polarization vector of compartment c (a length), H_c its access filter, and $\omega_c = 2\pi f_c$ [15, 26]. Equation (8) uses amplitude gains; relative phases can be retained in the

complex transfer. The passive compartment equation $C\dot{\mathbf{V}} + A\mathbf{V} = \mathbf{I}^0 - \mathbf{p}E$ [19] gives $\delta\mathbf{V}(\omega) = -(A + i\omega C)^{-1}\mathbf{p}E(\omega)$, with capacitance matrix C, conductance matrix A, endogenous current $\mathbf{I}^0$, and field-to-current coupling $\mathbf{p}$. The static gain and its frequency dependence belong to this one response. A model driven by post-coupling polarization and a model driven by current through its own membrane equation must therefore count that filtering once.

For a dominant entry point, $\varepsilon_j = \lambda_f(\omega_j)E_j$ denotes the polarization reaching the nonlinearity. The direct-coupling calculation takes $\lambda_f \approx \lambda_0$ to isolate mixing and recurrent amplification; §2.6 restores frequency-dependent access. A fast compartment could justify this approximation, but its identity and coupling are not inferred from the population model.

Expanding σ about an operating point v^* (a Volterra/Taylor expansion, as in [14]) gives $\sigma(v^*+s) \approx \sigma(v^*) + \sigma'(v^*)\,s + \frac{1}{2}\sigma''(v^*)\,s^2 + \dots$ The truncation requires $|s| \ll 1/\rho \approx 1.8$ mV and a nonzero second derivative for quadratic mixing to be leading. Larger excursions require the full transfer curve. The linear term preserves the two input frequencies. The quadratic cross term generates their difference through the identity $2\cos\omega_1 t\cos\omega_2 t = \cos\Omega t + \cos(\omega_1+\omega_2)t$. Averaging over the carrier (the fast $\cos^2\omega_j t$ terms average to $\frac{1}{2}$, the sum-frequency term to zero),

$$\big\langle \sigma(v^*+s)\big\rangle_c = \sigma(v^*) + \tfrac{1}{4}\sigma''(v^*)\,(\varepsilon_1^2+\varepsilon_2^2) + \tfrac{1}{2}\sigma''(v^*)\,\varepsilon_1\varepsilon_2\cos\Omega t + O(\varepsilon^4), \tag{9}$$

so a line appears at exactly the envelope frequency Ω, with amplitude

$$A_\Omega \simeq \tfrac{1}{2}\,\sigma''(v^*)\,\varepsilon_1\varepsilon_2, \tag{10}$$

where $\simeq$ means to leading order in the polarizations, here and in what follows. The same expansion leaves a constant (DC) term, $\frac{1}{4}\sigma''(\varepsilon_1^2+\varepsilon_2^2)$, which any read-out of the mean rate must include.

For a dynamic cellular response with synaptic input held fixed, define the effective second-order coefficient $\chi^{(2)}$ by the complex beat amplitude $\widehat{r}_\Omega$,

$$\widehat{r}_\Omega \simeq \chi^{(2)}(\omega_2,-\omega_1)\,\varepsilon_2\varepsilon_1, \qquad r_\Omega(t) = \mathrm{Re}\Big[\widehat{r}_\Omega e^{i\Omega t}\Big].$$

Here $r_\Omega(t)$ is the generated rate component before recurrent feedback. With this cosine-amplitude convention, the memoryless limit is $\chi^{(2)} = \sigma''(v^*)/2$. Multiple cellular processes can contribute to the same coefficient. Small amplitude justifies the second-order expansion; it does not remove memory, so the coefficient may depend on both frequencies and the operating state. Symmetry or cancellation can make it vanish. QIF and the sigmoid realize this shared quadratic structure with different coefficients.

What the quadratic term recovers is not the envelope but its square. Writing $A(t)$ for the instantaneous envelope of the two-tone field, $A^2(t) = \varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2\cos\Omega t$, the low-pass part of the squared signal is exactly

$$\mathrm{LPF}\big[s^2(t)\big] = \tfrac{1}{2}A^2(t) = \frac{\varepsilon_1^2+\varepsilon_2^2}{2} + \varepsilon_1\varepsilon_2\cos\Omega t. \tag{11}$$

This is square-law detection in the strict sense: the population transduces A^2, not A. The distinction matters because A^2 is exactly sinusoidal at Ω whereas A is not—at a matched target ($\varepsilon_1 = \varepsilon_2 = \varepsilon$), $A(t) = 2\varepsilon|\cos(\Omega t/2)|$, a rectified cosine.

The amplitude-modulated surrogate. The dense perturbative sweeps below use the three-line waveform

$$s(t) = \varepsilon\,[\,1 + m\cos\Omega t\,]\cos\omega_c t, \qquad \omega_c = 2\pi f_c, \tag{12}$$

with modulation depth m, because it parameterizes a sweep by one amplitude. It is not the physical signal: it carries a line at ω_c that the two-tone sum lacks. The two agree at Ω to leading order (the surrogate gives $A_\Omega = \frac{1}{2}\sigma''\varepsilon^2 m$) and are *not* equivalent beyond it. For matched genuine

tones LPF$[s^2] = \varepsilon^2(1 + \cos\Omega t)$, whereas for the $m = 1$ AM surrogate LPF$[s^2] = \varepsilon^2(\frac{3}{4} + \cos\Omega t + \frac{1}{4}\cos 2\Omega t)$: the same Ω amplitude, but a different DC term and a spurious second harmonic at 25% of the fundamental. Since that second harmonic is the observable that distinguishes transduction classes (§2.4), every claim that turns on the waveform is run with genuine two tones: the demodulation demonstration (Fig. 3), the curvature and square-law controls (Fig. 6), the laminar maps (Figs. S1, S7) and the kHz demonstration. The AM surrogate is retained for the dense perturbative sweeps in (Ω, p) and (Ω, C), which read only the leading Ω component, and Fig. 6d verifies that the two agree there.

In the language of [14], the quadratic cross-term $s_1 s_2$ between a slow and a fast input (the second-order Volterra kernel) implements amplitude modulation/demodulation and the associated intermodulation lines. Demodulation is thus itself an intermodulation product: the line at Ω is the *difference frequency* of the two applied tones (ω_1 and ω_2), generated by the same quadratic term that, acting again on an intrinsic rhythm ω_0, produces the $\omega_0 \pm \Omega$ sidebands seen at the spiking level (Fig. S12b). Equation (10) carries three predictions, all verified below: the recovered amplitude is quadratic in the drive ($A_\Omega \propto \varepsilon^2$, square-law detection); it follows the sigmoid curvature ($A_\Omega \propto \sigma''(v^*)$, hence zero at the inflection $v^* = v_0$); and, *at fixed post-coupling polarization* ε, it is independent of the carrier ω_c. This last prediction needs care: as a function of the *applied* field the response inherits the coupling $\varepsilon = \lambda_f E_0$, so the field required for a given effect rises with f_c (§2.6)—consistent with the experimental consensus. Carrier independence is thus a statement about the recovered beat component at fixed polarization, not about the applied field.

2.3 The exact mean field connects static and dynamic mixing

The adiabatic condition in §2.1 determines whether the static curve describes carrier mixing. Consider an uncoupled MPR population (Eq. (4), membrane time set to one), with constant drive Q and a weak added current $a\cos\omega t$. To second order in a, the carrier-averaged rate shift is $a^2 C(Q, \omega)$, where

$$C(Q,\omega) = \frac{r^*\,[\omega^2 + 3d^2 - \nu^2]}{(d^2+\nu^2)\{[d^2+\nu^2-\omega^2]^2 + 4d^2\omega^2\}}, \qquad r^* = F(Q), \quad d = \frac{\Delta}{\pi r^*}, \quad \nu = 2\pi r^*. \tag{13}$$

The fixed-point eigenvalues are $-d \pm i\nu$: d is the relaxation rate and ν the ringing angular frequency, both in membrane-time units. The limiting responses are

$$C(Q,\omega) \xrightarrow[\omega\to 0]{} \tfrac{1}{4}F''(Q), \qquad C(Q,\omega) \xrightarrow[\omega \gg d,\nu]{} \frac{F'(Q)}{2\omega^2}. \tag{14}$$

In the first limit the population tracks its static curve, whose curvature rectifies the input. In the second, the voltage excursion is approximately a/ω; its square supplies a slow drive that the population reads through F'. Above the inflection of F, these limits have opposite signs and the coefficient vanishes at $\omega_z = \sqrt{\nu^2 - 3d^2}$ (Fig. 7a). The 300 Hz and kHz carriers are on the fast side for the parameters of the pyramidal–interneuron network gamma (PING) circuit used below. That classification is parameter-dependent: a shorter effective membrane time can bring the same physical carrier into the adiabatic regime.

Membrane access is already included in this calculation. The linear voltage response is $H_v(\omega) = (i\omega + d)/[(i\omega + d)^2 + \nu^2]$. Dividing by its squared magnitude gives the mixing coefficient at fixed voltage excursion,

$$K(\omega) = \frac{C(Q,\omega)}{|H_v(\omega)|^2} = \frac{F'(Q)}{2}\,\frac{\omega^2 + 3d^2 - \nu^2}{\omega^2 + d^2}. \tag{15}$$

This coefficient tends to $F'/2$ at high frequency. The $1/\omega^2$ suppression at fixed current comes from $|H_v|^2$, the membrane filtering already represented by Eq. (8); it is not an additional detector filter. A static curve placed after the same membrane has the same leading carrier roll-off, although its

mixing coefficient and sign can differ. The steady-state curvature is $F'' = 4r^*(3d^2-\nu^2)/(d^2+\nu^2)^3$, so the inflection occurs at $\nu^2 = 3d^2$.

For the full eight-state PING (Eq. (27)), each high-frequency current tone of amplitude a_j produces a voltage phasor $a_j/(i\omega_j\tau_E)$. The v_E^2 term therefore supplies a difference-frequency source of magnitude $a_1a_2/(\tau_E^2\omega_1\omega_2)$ inside the equation for $\tau_E\dot{v}_E$. After division by τ_E, this source is propagated through the network susceptibility at Ω. At four stable operating points, the resulting response agrees with the full second-order calculation to within 1.1% at 300 Hz, 0.1% at 1 kHz, and 0.03% at 2 kHz (Fig. 7b).

The carrier and beat have different routes to the firing rate. At high frequency, the voltage equation integrates the input once, and the rate equation integrates the voltage again, giving a linear rate response proportional to $1/\omega^2$. The beat is generated in the voltage equation before this second integration and is read through the susceptibility at Ω. Thus its carrier dependence is $1/(\omega_1\omega_2)$, not the product of two linear rate responses. Among the quadratic terms in Eq. (4), v^2 contributes at order ω^{-2}, rv at ω^{-3}, and r^2 at ω^{-4} for nearby fast tones. Table 1 summarizes the comparison.

stage	static curve, same membrane	dynamic population
current → voltage	ω^{-1}	ω^{-1}
voltage → rate, carrier	1	ω^{-1}
current → rate, carrier	ω^{-1}	ω^{-2}
two tones → beat in the rate	$(\omega_1\omega_2)^{-1}$	$(\omega_1\omega_2)^{-1}\times$ const.

Table 1: Fast-carrier scaling with matched membrane access, at fixed beat frequency and a nonzero leading mixing coefficient. The dynamic population integrates the carrier once more between voltage and rate; the beat is formed before that step. Prefactors depend on the operating point and beat-frequency susceptibility.

2.4 Transduction class and its spatial and spectral signatures

A neural response proportional to the squared envelope is not equivalent to one proportional to the envelope. A detector implements some transduction $F(A)$ of the local envelope $A(t)$, and the two laws predict different amplitude scaling, harmonic content and spatial weighting.

For a narrowband field $s(t) = A(t)\cos\omega_c t$ passing through a memoryless nonlinearity g, the low-frequency (baseband) output Y_{LF} is the average of g over one carrier cycle:

$$Y_{\mathrm{LF}}(A) = (2\pi)^{-1}\int_0^{2\pi} g(v^* + A\cos\theta)\,\mathrm{d}\theta.$$

Odd powers of A average away, leaving only even ones:

$$Y_{\mathrm{LF}}(A) = g(v^*) + \tfrac{1}{4}g''(v^*)A^2 + \tfrac{1}{64}g^{(4)}(v^*)A^4 + \cdots \tag{16}$$

A smooth memoryless response with $g''(v^*) \neq 0$ therefore transduces A^2 at leading weak-input order, whatever its detailed shape. This includes the sigmoid mechanism of §2.2. Recovering A itself requires a qualitatively different detector: strong rectification. A full-wave rectifier gives $\mathrm{LPF}[\,|A\cos\omega_c t|\,] = (2/\pi)A$ and a half-wave rectifier $(1/\pi)A$, both linear in the envelope, as does a diode-and-capacitor peak detector. A hard threshold is not one of them: a step detector with offset δ transduces $\pi^{-1}\arccos(\delta/A)$ for $A > \delta$ and nothing below it, which is neither linear nor quadratic in A (App. D). Asymmetric channel rectification, threshold crossing and release events are candidate biological realizations of rectifiers of one kind or another. Under a linear downstream response and fixed coupling, the amplitude sweep below distinguishes a quadratic effective drive from one proportional to the envelope; the derivation of §2.2 assumes a smooth transfer, and this section asks what would show that assumption to fail.

The two classes are distinguishable at a single site. At a matched target, $A(t) = 2\varepsilon|\cos(\Omega t/2)|$, whose Fourier series $|\cos u| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{n\geq 1}\frac{(-1)^{n+1}}{4n^2-1}\cos 2nu$ gives a harmonic ladder in ratios

$A_{2\Omega}/A_\Omega = 1/5$, $A_{3\Omega}/A_\Omega = 3/35$ and $A_{4\Omega}/A_\Omega = 1/21$. The square-law output $\frac{1}{2}A^2 = \varepsilon^2(1+\cos\Omega t)$ contains only DC and Ω. For a mixed detector $F(A) = c_1 A + c_2 A^2$ the pre-network ratio is

$$r_{\mathrm{det}} \equiv \left|\frac{Y_{2\Omega}}{Y_\Omega}\right| = \frac{1/5}{1 + \frac{3\pi}{4}\, c_2\varepsilon/c_1}, \tag{17}$$

so measuring r_{det} estimates $c_2\varepsilon/c_1$ directly.

Two effects make the harmonic ratio usable only with care, and the second decides how to run the test. First, a smooth detector does generate a second harmonic beyond quadratic order, $Y_{2\Omega}/Y_\Omega \sim (c_4/c_2)\varepsilon^2$, so the square-law signature is a *vanishing* ratio in the weak-field limit, not an exact zero. Second, and more seriously, the recurrent network multiplies each harmonic by its own transfer gain, $r_{\mathrm{meas}} = r_{\mathrm{det}}\,|G(2\Omega)/G(\Omega)|$. A resonant column suppresses the second harmonic hard: driven at $\Omega = \omega_0$ with $\mathcal{Q} = 10$, a true 20% ratio reads as 0.7%, enough to make an envelope follower look square-law. The resonance is thus what would hide the discriminator, so the test must be run *off* resonance: at $\Delta f \simeq f_0/4$ both lines sit on the flat part of G, where $|G(2\Omega)/G(\Omega)| \approx 1.22$–$1.25$ across $\mathcal{Q} = 3$–30, or with G measured independently by a weak sweep of direct low-frequency stimulation (transcranial alternating-current stimulation, tACS) and deconvolved.

An amplitude sweep tests the scaling rather than one ratio. An envelope-following detector gives $Y_\Omega \propto \varepsilon$ with the harmonic ratio pinned at 1/5; a weak square-law detector gives $Y_\Omega \propto \varepsilon^2$ with a ratio growing as ε^2. The log–log slope of the envelope-locked response against field amplitude therefore separates the classes without requiring absolute calibration; the ε^2 control of §4.4 is that measurement, performed in the model.

Beyond harmonics and amplitude scaling, the transduction class fixes the spatial map. Define the excursion as $\mathcal{M}[y] \equiv \max_t y - \min_t y$ of a periodic quantity $y(t)$. The conventional TI envelope map is $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$, whereas the square-law detector responds to $\mathcal{M}[\frac{1}{2}A^2] = 2\varepsilon_1\varepsilon_2$, so the two are related pointwise by

$$\mathcal{M}\left[\tfrac{1}{2}A^2\right] = \max(\varepsilon_1, \varepsilon_2)\; \mathcal{M}[A]. \tag{18}$$

The square-law map is the conventional envelope-excursion map multiplied, at each point, by the dominant local polarization. It therefore does not preserve conventional TI focality. In typical montages, where the dominant carrier is substantially larger superficially than at depth, this reweighting degrades deep-to-superficial contrast: at $(\varepsilon_1, \varepsilon_2) = (10, 1)$ superficially against $(0.5, 0.5)$ at a matched deep target, $\mathcal{M}[A]$ favors the superficial point by 2× and $\mathcal{M}[\frac{1}{2}A^2]$ by 40×. The magnitude of the degradation is not fixed by this algebra and must be evaluated from the actual vector field distribution; a sufficiently weak second carrier can suppress the superficial product instead. The product law is not an account of TI focality.

The selectivity this mechanism does supply is dynamical rather than spatial. Writing $R_\Omega(\mathbf{x})$ for the demodulated response at the envelope frequency Ω produced at location $\mathbf{x}$, and $G_{\mathbf{x}}(\Omega)$ for the local network susceptibility there,

$$R_\Omega(\mathbf{x}) \;\propto\; \sigma''(v^*_{\mathbf{x}})\; G_{\mathbf{x}}(\Omega)\; \varepsilon_1(\mathbf{x})\, \varepsilon_2(\mathbf{x}), \tag{19}$$

a product of three factors. Field overlap sets where a difference-frequency drive is generated at all; the operating point and the network susceptibility set which of the driven populations amplifies it. Resonance does not override an unfavorable field product, but it can compensate for one. This is the radio analogy taken literally: received signal strength times receiver tuning.

Equation (19) is written for scalar projected fields. In general the coupling is vectorial and compartment-specific, $\varepsilon_i(\mathbf{x}) = \boldsymbol{\lambda}\cdot\mathbf{E}_i(\mathbf{x})$, so the mixing product $(\boldsymbol{\lambda}\cdot\mathbf{E}_1)(\boldsymbol{\lambda}\cdot\mathbf{E}_2)$ is signed: its magnitude follows the product of the projected amplitudes, while its phase inverts with either the sign of σ'' or the relative sign of the two projections. The derivations below take the collinear, scalar case.

2.5 Amplification: the synaptic low-pass and the network resonance

Synaptic filtering suppresses the carrier after the nonlinearity has generated the beat. Each JR excitatory synapse has transfer function $H_s(\omega) = Aa/(i\omega + a)^2$, a second-order low-pass with characteristic frequency $a/(2\pi) \approx 16$ Hz. It passes a 10 Hz component with relative amplitude about 0.7, while reducing a 100 Hz component to about 2.5% (Fig. 5). Provided the carrier does not itself excite a responsive mode, the loop therefore reads predominantly the generated low-frequency drive.

What turns this recovered beat component into a large, frequency-selective response is the network. Linearizing the mass about its fixed point gives a leading eigenvalue pair $\mu_\pm = -\gamma \pm i\omega_0$; near the Hopf the response is dominated by this critical mode, so the whole network behaves as a single weakly damped harmonic oscillator with damping γ and natural frequency ω_0 (the gain below is its one-mode, center-manifold reduction). Below a supercritical Hopf, $\gamma \to 0^+$ and the mass is a sharply tuned (high quality-factor $\mathcal{Q} = \omega_0/2\gamma$) resonator; driving it with the demodulated beat component (10) at Ω gives the resonance gain of a damped linear oscillator,

$$G(\Omega) \propto \left[(\omega_0^2 - \Omega^2)^2 + (2\gamma\Omega)^2\right]^{-1/2}, \qquad G(\omega_0) \propto \frac{1}{2\gamma\omega_0}, \tag{20}$$

so the steady response $\sim A_\Omega\, G(\Omega)$ peaks at $\Omega = \omega_0$ (a Lorentzian in power) with a height that diverges as the Hopf is approached, within the weak-drive regime where the linear response holds. Here ω_0 is the imaginary part of the eigenvalue; the undamped frequency of the canonical oscillator differs from it by a term of order γ^2, immaterial near the Hopf. Above the Hopf the mass is an autonomous limit cycle at ω_0, and the demodulated drive *entrains* it (an Arnold tongue), giving a response peak near ω_0 that grows with cycle amplitude.

Collecting the stages, the response at the beat is

$$R_\Omega \simeq G(\Omega;\gamma)\; D\; \lambda_f(\omega_1)E_1\, \lambda_f(\omega_2)E_2, \tag{21}$$

where λ_f maps field to local carrier polarization, D converts the product of those polarizations into a beat drive, and G maps that drive to the observed response. Their normalization depends on where the generated drive enters. For JR, $D = \frac{1}{2}\sigma''(v^*)$ and G is the response to a generated rate drive. For fast-carrier QIF in the normalized units of Eq. (4), $D = 1$ for the v^2 source and G includes its conversion through the voltage and rate dynamics. Thus the population readout is included once. In these limits D is independent of carrier frequency at fixed polarization; across the dynamic crossover it acquires frequency dependence. A carrier sweep and a beat sweep therefore constrain different stages rather than a single effective gain.

The resonance does not supply the quadratic mixer. The cubic Hopf normal form responds to an in-band drive but has no second-order difference-frequency source (App. C). A large resonant susceptibility alone therefore does not establish beat generation.

2.6 Carrier access depends on the effective membrane response

Carrier independence at fixed polarization does not imply independence at fixed applied field. To quantify access, consider a first-order membrane response $H_m(\omega) = 1/(1 + i\omega\tau_m)$, with effective time constant τ_m. For nearby carriers, the two polarizations acquire nearly the same attenuation. The quadratic beat then satisfies

$$\frac{A_\Omega(f_c)}{A_\Omega^{\text{direct}}} \simeq |H_m(\omega_c)|^2 = \frac{1}{1 + (\omega_c\tau_m)^2} \sim \frac{1}{(\omega_c\tau_m)^2} \quad (\omega_c\tau_m \gg 1), \tag{22}$$

where A_Ω^{direct} is the beat generated with the same static coupling and no access attenuation. This filter belongs either in the field-to-polarization map or in the membrane equation that computes that polarization.

Somatic passive time constants are commonly 10–30 ms; reported human pyramidal values include approximately 12 ms [27] and 16.5 ± 3.7 ms [28]. For the illustrative $\tau_m = 16$ ms used here, the power factor $|H_m|^2$ is about 10^{-4} at 1 kHz, 2.5×10^{-5} at 2 kHz, and 10^{-6} at 10 kHz. A static nonlinearity placed after this somatic response therefore receives strongly attenuated carriers.

A shorter effective time changes this conclusion quantitatively. The sensitivity case $\tau_m = 0.2$ ms gives $|H_m|^2 \approx 0.39$ at 1 kHz and 0.14 at 2 kHz (Fig. S5). These are assumed access parameters, not measured TI parameters of the AIS or a node. They can represent a faster compartment in Eq. (8), or motivate a shorter membrane time in a QIF reduction. In the latter description, the approach to a static transfer is governed by Eq. (6); the membrane filter must not be applied again outside the QIF equations. A short access time improves carrier transmission without, by itself, specifying the mixing coefficient or the remaining circuit resonance.

Element	Input	Illustrative corner	Interpretation
JR excitatory synapse	rate r	$\sim$16 Hz	downstream synaptic filter
somatic access	field E	$\sim$10 Hz	$\tau_m = 16$ ms example
fast effective access	field E	$\sim$800 Hz	assumed $\tau_m = 0.2$ ms
channel kinetics	local voltage	model-dependent	part of the dynamic detector

Table 2: Distinct response stages. The access corners illustrate specified time constants; the fast row does not identify an anatomical compartment. Synaptic filtering follows beat generation, whereas access controls the carrier reaching the nonlinearity.

Geometry can also alter the relevant response mode. For a passive cable, the magnitude of the frequency-dependent space constant is

$$\ell(\omega) = \ell_0(1 + \omega^2\tau_m^2)^{-1/4}, \tag{23}$$

where ℓ_0 is its static value. At 1 kHz and $\tau_m = 16$ ms, this length is about one tenth of its static value, so compartments that are strongly coupled at low frequency can respond differently to the carrier. A somatic charging time alone therefore does not determine every extracellular-field response mode. The periaxonal and field-charging evidence is considered in §5.

Channel kinetics contribute to the effective dynamic mixing response. An additional gate filter would duplicate any dynamics already included there; neither a static sigmoid nor QIF specifies a universal 1 kHz gate cutoff. Equation (13) gives the QIF realization of this response. Carrier sweeps can constrain the combined access and mixing response, while beat-frequency sweeps probe the downstream amplifier.

2.7 Operating-point dependence

Because $A_\Omega \propto \sigma''(v^*)$, demodulation is governed by the curvature at the operating point, and in particular by how far v^* sits from the sigmoid inflection (Fig. 6). The Population Excitation–Inhibition Index (PEIX) of [13] is a related but distinct quantity: the sigmoid gain $\sigma'(v^*)$ normalized to its maximum $\sigma'(v_0) = e_0\rho/2$ at the inflection,

$$\mathrm{PEIX}(v^*) \equiv \frac{\sigma'(v^*)}{\sigma'(v_0)} = \mathrm{sech}^2\Big[\tfrac{\rho}{2}\,(v^* - v_0)\Big] \in (0, 1], \tag{24}$$

a dimensionless read-out of where v^* sits, the side given by $\mathrm{sign}(v^* - v_0)$. PEIX thus tracks σ' (maximal at the inflection v_0), whereas the demodulation gain tracks σ'' (zero at v_0, peaked on the flanks, and sign-reversing across it): the two are complementary, and PEIX peaks where the demodulation gain vanishes.

This complementarity is a property of the memoryless detector. In the exact mean field at a fast carrier the rectified DC term follows the slope F' and is largest at the inflection (§2.3), so the relation between PEIX and TI efficacy is model-dependent; PEIX remains a measurable read-out of the operating point. In the Jansen–Rit detector, anything shifting v^* (neuromodulation, attention, arousal, or pathology such as psychedelics or PV-interneuron dysfunction in Alzheimer's disease) scales, and can flip the sign of, the TI response; in the dynamic detector the same shifts act through F' and the susceptibility, without a sign rule. The state dependence is the falsifiable handle; the sign reversal is specific to the memoryless model.

3 Methods

3.1 Single-column Jansen–Rit: integration, lock-in, and controls

All symbols and units are collected in Table 1 (App. A). We integrate the single-column JR model of §2.1 (Eqs. (1), parameters in Table 2) as six first-order ODEs. The TI field enters only the pyramidal output sigmoid, Eq. (1), as $s(t)$; the read-out $v = y_1 - y_2$ is *not* fed the field directly, so any power at Ω in v is produced by the nonlinearity and the recurrent loop, not by trivial pass-through. Unless stated otherwise $f_c = 100$ Hz and $m = 1$.

We integrate with a fixed-step fourth-order Runge–Kutta (RK4) scheme, $\Delta t = 2 \times 10^{-4}$ s (50 samples per carrier cycle at 100 Hz; reduced to 10^{-4} s for carrier sweeps up to 300 Hz). The kHz-carrier runs of Fig. S9 use $\Delta t = 2 \times 10^{-5}$ s for the carrier sweep to 4 kHz (25 samples per carrier cycle at 2 kHz, 12.5 at 4 kHz) and 1.5×10^{-5} s for the 2 kHz output spectrum. Halving each step changes the reported lock-in amplitudes by less than 1%. The 10 kHz figures quoted in §2.6 are analytic evaluations of Eq. (22), not simulations.

For each measurement the system is settled for 10–16 s (longer near the Hopf, where damping is weak) before a 3–20 s measurement window; the resonance sweeps use a 20 s window so the lock-in averages the limit-cycle beating on the entrainment side. Parameter sweeps are vectorized: the state is an $(N, 6)$ array advanced simultaneously for all N parameter combinations, so a full resonance map runs in seconds.

The full bifurcation diagram (Fig. 2) is computed by numerical continuation of the fixed-point and limit-cycle branches. We independently locate the operating supercritical Hopf by solving, for each p, the self-consistent scalar equation for the resting fixed point $v = y_1 - y_2$, forming the 6×6 Jacobian numerically and diagonalizing it: the Hopf is the p at which the leading eigenvalue pair crosses the imaginary axis with $\mathrm{Im}\,\mu_+$ in the alpha band. This gives $p \approx 315$ Hz and $f_0 = \mathrm{Im}\,\mu_+/2\pi \approx 11.1$ Hz—consistent with the continuation and with the JR bifurcation analysis of [29]. For $p > 315$ the fixed point is a stable focus (resonator); between the saddle-node on an invariant circle (SNIC, $p \approx 114$) and the Hopf the column free-runs on the alpha limit cycle.

The response at the envelope frequency is read out with a software lock-in amplifier referenced to Ω—a standard way to isolate the component of a signal at one chosen frequency: multiply by sine and cosine at Ω and average, so everything *not* at Ω cancels and only the envelope-locked amplitude survives. We multiply $v(t)$ by quadrature references and integrate over a Hann-windowed measurement window to obtain the in-phase (I) and quadrature (Q) components,

$$I = \frac{2}{\sum_k w_k} \sum_k w_k \,(v_k - \bar{v}) \cos \Omega t_k, \quad Q = \frac{2}{\sum_k w_k} \sum_k w_k \,(v_k - \bar{v}) \sin \Omega t_k, \quad A_\Omega = \sqrt{I^2 + Q^2}, \tag{25}$$

with w a Hann window and $\bar{v} = \sum_k w_k v_k / \sum_k w_k$ the windowed mean. For a pure tone $v = A\cos(\Omega t + \phi)$ the product-to-sum identity gives $I \to A\cos\phi$ and $Q \to -A\sin\phi$, so $A_\Omega \to A$; the factor $2/\sum_k w_k$ is the normalization that returns the physical amplitude, and A_Ω equals twice the magnitude of the windowed Fourier coefficient at Ω. Out-of-band components can leak into the estimate, particularly the DC level of v ($\bar{v} \sim 8$ mV, far larger than the demodulated signal). Over

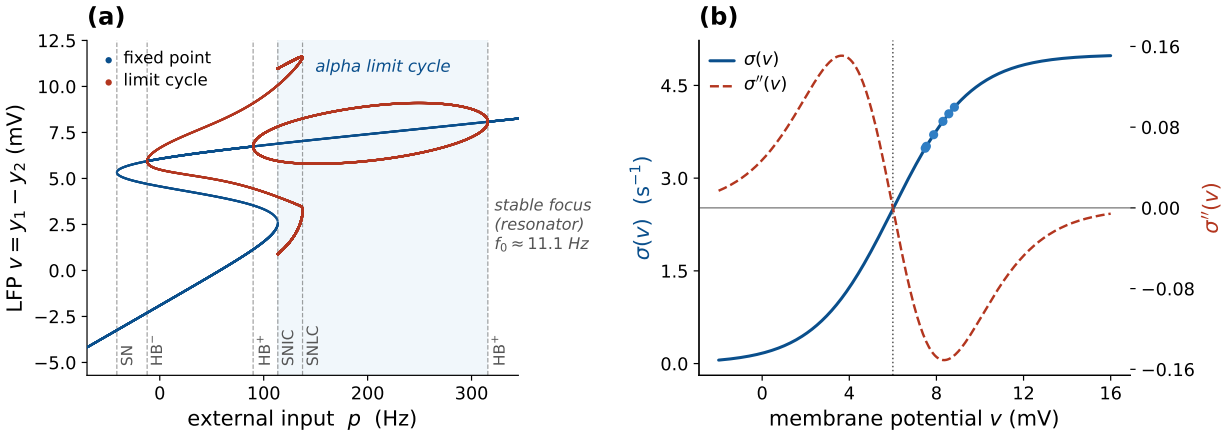


Figure 2: The Jansen–Rit operating regime: bifurcation and sigmoid curvature. *(a)* JR bifurcation diagram from numerical continuation—the LFP $v = y_1 - y_2$ along the fixed-point branch (blue) and the min/max of the limit-cycle branches (red) vs. external input p, with *solid* stable and *faded* unstable. The alpha limit cycle (shaded) is bounded by a saddle-node-on-invariant-circle (SNIC, $p \approx 114$) and the supercritical Hopf ($p \approx 315$, $f_0 \approx 11.1$ Hz); beyond the Hopf the fixed point is a stable focus, the near-Hopf resonator used here. Further codim-1 points are marked (saddle-node SN, Hopf HB$^{\pm}$, saddle-node of limit cycles SNLC). *(b)* The sigmoid $\sigma(v)$ and its curvature $\sigma''(v)$, which sets the demodulation gain and vanishes at the inflection v_0. Dots mark the operating points reached by the closed-loop model for the p values used here.

an exact integer number of Ω-periods $\sum_k w_k \cos \Omega t_k = 0$ and there is no leakage, but in general it is nonzero, so we subtract $\bar{v}$ explicitly; this makes the metric accurate down to the 10^{-8} level needed for the kHz analysis (§2.6). The signed in-phase component I alone recovers $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ including its sign, which we use to verify the curvature law (Fig. 6).

We compare the *open-loop detector*, the firing-rate sigmoid at a fixed operating point v^* without recurrence, with the *closed-loop* column, the full recurrent Jansen–Rit with its feedback. The open-loop detector passes the two-tone field through σ at v^* and applies Eq. (25), isolating the pure demodulation law (Figs. 5, 6, S5). The *bifurcation diagram* records the field-free min/max of v vs. p, each run initialized at its fixed point plus a small kick. The *two-dimensional map* evaluates A_Ω over an (Ω, p) grid. We verify the mechanism three ways: the square law (A_Ω vs. ε on a well-damped focus, expecting a log–log slope of 2); a linearization control (replacing σ in Eq. (1) by its tangent at v^*, which must abolish the Ω response); and a check of σ', σ'' against finite differences. All code is openly available (see *Code and data availability*).

3.2 Laminar two-band model (LaNMM)

We complement the single-column analysis with simulations of the laminar neural mass model (LaNMM) of §2.1. We integrate the full 28-variable column [13, 25], with the field entering as in the single JR column, as a membrane polarization added inside the sigmoid argument of the target population,

$$\varphi_X(t) = \sigma\big(v_X(t) + s(t)\big), \qquad X \in \{P_1, P_2\}, \tag{26}$$

with $s(t)$ the AM or two-tone field of §2.2 (amplitude ε in mV, beat $\Omega = 2\pi\Delta f$, carrier $\omega_c = 2\pi f_c$) and all three external drives held flat at their baselines (Fig. S2). A perturbation of an external input *rate* would require clipping at zero once its excursion exceeds the baseline, and a hard clip is a half-wave rectifier—a diode detector placed upstream of the sigmoid, which would inject a line at Δf into the drive before the population nonlinearity received it. A signed membrane polarization carries no such constraint and none is applied. Because the LaNMM and JR sigmoids share the voltage scale $1/\rho \approx 1.8$ mV, ε is directly comparable between the two models.

Integration is a vectorized fixed-step RK4 ($\Delta t = 10^{-4}$ s) advancing the whole parameter grid

simultaneously, as in the JR sweeps. In place of the lock-in (Eq. (25)) we read the alpha-band (8–12 Hz) power of the P_1 and P_2 membrane potentials as the mean squared magnitude of the Hann-windowed fast Fourier transform (FFT) over the band, in arbitrary units, over a post-transient window, a phase-agnostic measure suited to the entrainment regime, where lock-in phase slips would otherwise complicate the read-out.

Two sweeps probe the column: an amplitude–beat sweep over $(\varepsilon, \Delta f)$ at fixed carrier (the Arnold tongue), and a carrier–beat sweep over $(f_c, \Delta f)$ at fixed ε, built on the LaNMM module of [13]. The maps (Figs. S1, S7) show the *signed* stimulation-induced change, alpha power minus the unstimulated ($\varepsilon = 0$) baseline, on a zero-centered scale; over most of the P_1-driven grids the change is negative. For the alpha resonance we instead drive the deep P_1 loop directly and sweep its input against Δf, reading the lock-in (Eq. (25)) of v_{P_1} at Δf.

3.3 Exact next-generation mean field (NMM2 PING)

We test the mechanism in the exact next-generation mean field (NMM2, §2.1), following the Rosetta Stone formulation of [15] (which places the Montbrió–Pazó–Roxin reduction [18] on a common footing with the JR and LaNMM models used here), where the firing-rate nonlinearity is the derived MPR dynamics of Eq. (4) rather than a postulated sigmoid. We use a two-population excitatory–inhibitory (PING) motif [19, 20, 15]: each population $\alpha \in \{E, I\}$ is an MPR mean field (firing rate r_α, mean potential v_α) with membrane time τ_α, coupled through second-order synapses,

$$\begin{aligned} \tau_E \dot{r}_E &= \Delta/(\pi\tau_E) + 2 r_E v_E, & \tau_E \dot{v}_E &= v_E^2 + \bar{\eta} - (\pi\tau_E r_E)^2 + \tau_E C\,(s_E - s_I) + I_E(t), \\ \tau_I \dot{r}_I &= \Delta/(\pi\tau_I) + 2 r_I v_I, & \tau_I \dot{v}_I &= v_I^2 + \bar{\eta} - (\pi\tau_I r_I)^2 + \tau_I C\,(s_E - 2 s_I), \end{aligned} \tag{27}$$

with second-order synaptic activations $s_E = \hat{K}_a[r_E]$ (excitatory, time constant τ_a) and $s_I = \hat{K}_g[r_I]$ (inhibitory, τ_g), each obeying the same critically-damped operator as the JR synapse, $\tau^2 \ddot{s} + 2\tau \dot{s} + s = r$; here C is the global synaptic gain and the weights $(A_{ee}, A_{ei}, A_{ie}, A_{ii}) = (1, 1, 1, 2)$ set the E and I rows: the E membrane sees $C\,(s_E - s_I)$ and the I membrane $C\,(s_E - 2s_I)$, the only asymmetry being the doubled $I \to I$ weight $A_{ii} = 2$.

The common background drive $\bar{\eta}$ is the bifurcation parameter: raising it past a supercritical Hopf at $\bar{\eta}_H = 1.110$ opens a gamma limit cycle (the gamma counterpart of the JR alpha Hopf). The applied field enters the excitatory voltage equation as a current-like forcing, $I_E(t) = P\,E(t)$ in the convention of the exact mean field [19], with the dipole coupling P instantaneous, as two genuine tones at $f_c \mp \Delta f/2$ (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) of amplitude A_f in the dimensionless MPR units of v_E. The E membrane $\tau_E \dot{v}_E$ integrates this forcing into the carrier-frequency excursion $A_f/(\omega\tau_E)$ that the exact quadratic v_E^2 term rectifies, with no postulated sigmoid; the PING as run therefore computes membrane access with $\tau_m = \tau_E$.

We integrate with fixed-step RK4 ($\Delta t = 0.05$ ms) and read out the lock-in (Eq. (25)) of r_E at Δf over the $(\Delta f, \bar{\eta})$ plane. Parameters (Rosetta Stone PING [15]): $\Delta = 1$, global coupling $C = 15$, $\tau_E = 15$, $\tau_I = 7.5$, $\tau_a = 10$, $\tau_g = 2.5$ ms, with $\bar{\eta}$ swept through the supercritical Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz), located by bisection on the leading eigenvalue of the mean-field Jacobian. The simulations retain these finite membrane times; Eq. (6) characterizes the fast-membrane limit.

Two parameters reach this gamma Hopf, and we use both as criticality knobs—just as the external drive p and the coupling C both reach the JR alpha Hopf. The background drive $\bar{\eta}$ (above) is swept for the resonance map (Fig. S8); the global coupling $C\,(\equiv J)$ is swept for the J-curve (Fig. S10): we vary J at fixed $\bar{\eta}$ toward the gamma Hopf J^* (located by bisection on the leading Jacobian eigenvalue) and read the lock-in peak of r_E over Δf. Unlike the JR J-curve, *no operating-point pinning is needed*: at this carrier the generated difference-frequency source is operating-point independent to about 1% (§2.3), so sweeping J changes the damping γ and the modal projections but not the source, and the gain is read as the response per unit source.

3.4 QIF spiking network (microscale of NMM2)

NMM2 is the exact mean field of an all-to-all network of QIF neurons [18, 15, 20], so we integrate that network directly to test the prediction at the spike level (§4.8): $N = 8000$ QIF neurons per population, parameters identical to the PING mean field (Eq. (27)), excitabilities $\eta_j \sim \text{Lorentzian}(\bar{\eta}, \Delta)$ sampled by deterministic quantiles, second-order synapses driven by the empirical population rates, and a genuine two-tone field entering the excitatory membranes. Spiking and threshold dynamics are where the AM surrogate and the physical waveform cannot be assumed equivalent beyond perturbation theory, so this run uses two tones throughout.

The bifurcation parameter is the common background drive $\bar{\eta}$, with the gamma Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz) from the mean-field Jacobian; the spiking network shows the same threshold and an autonomous $f_0 \approx 53$ Hz, with no finite-size offset to explain. The carrier is $f_c = 300$ Hz and the field amplitude $A_f = 35$. Because this is a *gamma* circuit, the TI beat Δf is set in the gamma band: $\Delta f = 55$ Hz *near* the forced gamma resonance ($\bar{\eta} = 0.85$, below the Hopf) and $\Delta f = 42$ Hz in the oscillatory regime ($\bar{\eta} = 1.5$, above the Hopf)—a detuning of $\approx$11 Hz below the autonomous $f_0 \approx 53$ Hz that places the beat outside the 1:1 Arnold tongue, so the above-Hopf response is gating, not entrainment (§F).

The network is integrated with Euler steps of 0.0025 ms, settled for 800 ms (over four relaxation times at $\bar{\eta} = 0.85$, where the mean-field relaxation time is 176 ms) and measured over an integer number of beat cycles, 44 at 55 Hz and 34 at 42 Hz (800 and 810 ms), so that the finite-window bias of phase statistics vanishes. Envelope-locked timing is quantified by the quadrature lock-in A_Ω of $r_E(t)$ at Δf, reported as the modulation depth $A_\Omega / \langle r_E \rangle$ (a stable measure; the off→on lock-in ratio would divide by the field-free floor), and per neuron by the pairwise phase consistency of each neuron's spike phases at Δf [30], which is unbiased in the spike count, over neurons with at least ten spikes in the window.

Halving the time step changes the lock-in and the pooled vector strength by under 1%; doubling the spike-detection threshold changes the lock-in by about 1% and the mean rate by 2%, but the pooled vector strength in the forced regime by about 45%, so phase concentration, unlike rate modulation, is sensitive to how spike times are defined.

3.5 Coupling sweeps, timing, and the tACS control (Jansen–Rit)

The coupling sweeps of Figs. 9 and S11, and the transcranial alternating-current stimulation (tACS) control of Fig. S3, use the single JR column with the operating point v^* pinned: for each global connectivity C we solve the field-free fixed-point relation analytically for the external drive $p(C)$ that holds v^* (hence $\sigma''(v^*)$) fixed, vary $C\,(= J)$ toward the alpha Hopf, and read γ, ω_0 from the 6×6 Jacobian. The field is the AM surrogate of Eq. (12) for TI (Fig. 9; AC lock-in vs DC mean rate for Fig. S11) or a direct in-band sinusoid $s(t) = \varepsilon \cos \Omega t$ ($\varepsilon = 0.1$ mV) for the tACS control (Fig. S3), each read out by the lock-in of v at Δf.

3.6 Vocabulary: forced response, entrainment, and gating

"Entrainment" is used in several senses, so we fix three terms. *Forced response* is the driven, non-autonomous case below the Hopf, quantified by the lock-in amplitude at the beat Δf (Eq. (25)). *Entrainment* is reserved for its dynamical-systems meaning, frequency capture of a *self-sustained* oscillator [31]; in the maps we screen for it with the dominance criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz on the dominant output frequency, a necessary but not sufficient condition for capture. *Gating* is amplitude modulation of a persisting rhythm without frequency capture [32, 33]. Capture and gating are dynamically different, and the same word would otherwise cover both; Appendix F gives the metrics and the Arnold-tongue protocol.

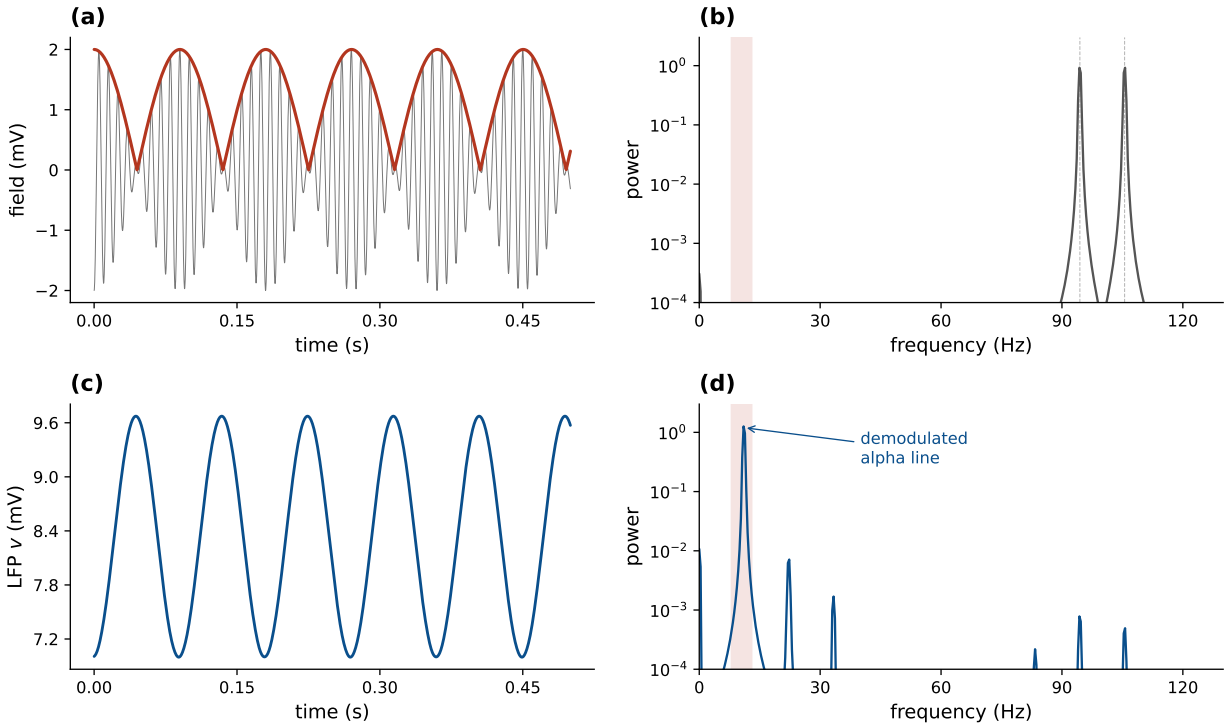


Figure 3: An alpha rhythm is synthesized from a two-tone TI input (Jansen–Rit below Hopf). The field is the physical TI waveform, two equal tones at $f_{1,2} = f_c \mp \Delta f/2$, not the amplitude-modulated surrogate. *(a)* The field $s(t)$ and its envelope $A(t) = 2\varepsilon|\cos(\Omega t/2)|$—a rectified cosine, not a sinusoid; it is A^2 that is sinusoidal at Ω (§2.2). *(b)* The input amplitude spectrum has lines only at f_1 and f_2 (94.45 and 105.55 Hz, $\Delta f = 11.1$ Hz) and no third line at the carrier; in the alpha band (shaded) it is 1.6×10^{-8} of the tone amplitude, finite-window leakage rather than physical power. *(c)* The output $v = y_1 - y_2$ oscillates at the envelope frequency. *(d)* The output spectrum has a strong peak at the beat, 11.1 Hz, synthesized by the sigmoid and absent from the input.

4 Results

Two rungs of the modeling ladder carry the evidence: the single Jansen–Rit column (closed form, memoryless detector) and the exact next-generation mean field (NMM2, the nonlinearity derived rather than postulated) with its spiking network. Each shows the same pair, quadratic difference-frequency generation followed by near-Hopf amplification, with different detector coefficients. Two mesoscale signatures follow: the network's own coupling sets the gain at fixed generated drive (the J-curve, §4.6), and the resonance amplifies the envelope-locked modulation far more than the mean rate (§4.7), which the QIF network tests at the spike level. The laminar two-band column, at carriers of 25–80 Hz and drives up to 3 mV, is an extension (App. E).

4.1 The Jansen–Rit sigmoid demodulates: an alpha line from a carrier-only input

Figure 2 orients the system: a supercritical Hopf at $p \approx 315$ with $f_0 \approx 11.1$ Hz, and a sigmoid whose curvature peaks well off the inflection, so the closed-loop operating points sit where σ'' is large. An input with no spectral power in the alpha band, with lines only at the two tones $f_c \mp \Delta f/2$, produces an output oscillation at the 11.1 Hz envelope frequency and an alpha peak in the output spectrum that is absent from the input (Fig. 3).

4.2 Near the Hopf, the demodulated beat drives a sharp resonance

The resonance is quantified in Fig. 4. In the fixed-point regime ($p > 315$, a stable focus) the response is a Lorentzian centered at f_0 whose height grows monotonically as the Hopf is approached ($0.23 \to 0.43 \to 0.70$ mV; Fig. 4), i.e. gain $\propto 1/\gamma$. This is forced gain: with no drive the focus is

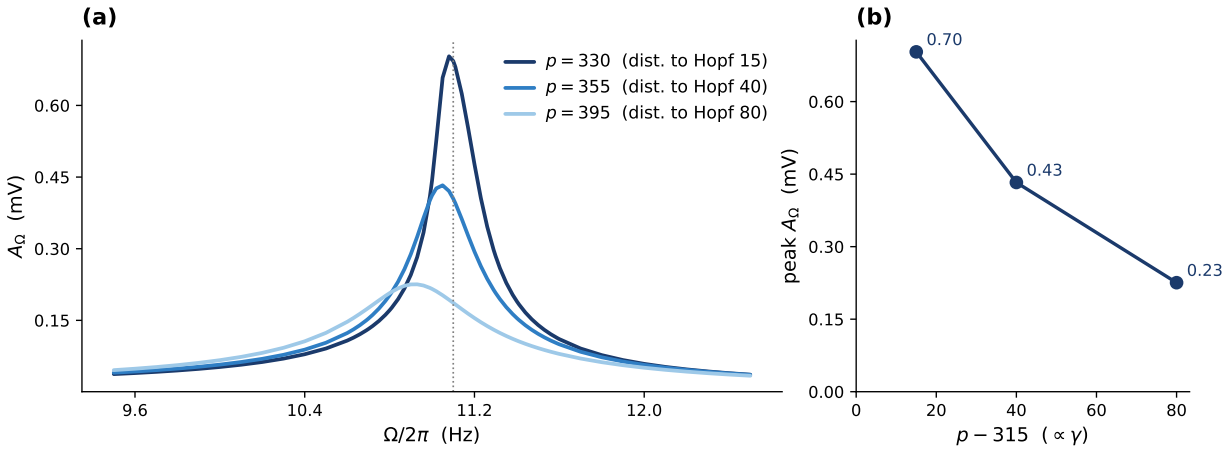


Figure 4: The demodulated beat drives a resonance that sharpens toward the Hopf (Jansen–Rit). Fixed-point regime ($p > 315$, stable focus): lock-in response at Ω vs. envelope frequency ($\varepsilon = 0.3$ mV). *(a)* The demodulated drive forces a Lorentzian resonance at $f_0 \approx 11.1$ Hz whose peak redshifts slightly as the damping grows ($\omega_{\text{peak}} = \sqrt{\omega_0^2 - 2\gamma^2}$; $11.08 \to 10.93$ Hz across the three curves); the response carries no power at Ω in the absence of drive, so this is forced gain. *(b)* The peak height grows as the Hopf is approached ($p \to 315$, distance to Hopf $\to 0$). Three points cannot establish a power law; the $1/\gamma$ scaling is quantified over a decade, together with its saturation, in Fig. 9(c). The above-Hopf regime, where the demodulated drive entrains the autonomous alpha, is characterized by output frequency in Fig. S4.

silent at Ω, so all power at Ω is synthesized by the demodulating nonlinearity and amplified by the recurrent loop. Above the Hopf ($p < 315$) the column free-runs at f_0 and the demodulated drive instead phase-locks that autonomous alpha within a 1:1 Arnold tongue; because lock-in amplitude at Ω then conflates the driven response with the free-running cycle, we characterize that regime with the output frequency f_{out} rather than the lock-in amplitude (Fig. S4).

4.3 Carrier independence and the kHz roll-off

At fixed post-coupling polarization the demodulated response is flat across the carrier band, for both the open-loop detector and the closed-loop column near the Hopf (Fig. 5). As a function of the *applied* field, by contrast, the membrane low-pass makes the response roll off as $1/f_c^2$ above the corner (Fig. S5): passive somatic demodulation is suppressed four to six orders of magnitude at kHz in the 16 ms access model, whereas the assumed 0.2 ms response retains appreciable carrier access.

We also drive the full recurrent column with a kHz field, passing it through an explicit membrane low-pass (an extra first-order state, time constant τ) for each coupling route before it reaches the sigmoid. The column demodulates and resonates at the recovered alpha in the fast-access case ($\tau = 0.2$ ms), as Eq. (22) predicts (Fig. S9a). Driving the fast-element route with two *genuine* tones at $f_{1,2} = 1994.54$ and 2005.46 Hz, centered near 2 kHz and beating at 10.91 Hz, the input carries 2.2×10^{-13} of its spectral amplitude in the 8–14 Hz band, i.e. none, while the output carries a line at the beat (Fig. S6), with the attenuation set by the analytic $|H_m|^2 = 0.137$ at 2 kHz for $\tau = 0.2$ ms. The kHz claim therefore rests on the two-tone waveform under the specified access assumption. The anatomical route is not determined by this calculation. Limited sensitivity to unmodulated kHz fields has been reported in hippocampal network oscillations [34].

4.4 The nonlinearity is the detector: the σ'' curvature law and its controls

The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the operating range, vanishing and reversing sign at the inflection v_0 (Fig. 6)—demodulation is the sigmoid-curvature term of Eq. (10), and the closed-loop operating points sit where σ'' is large and negative.

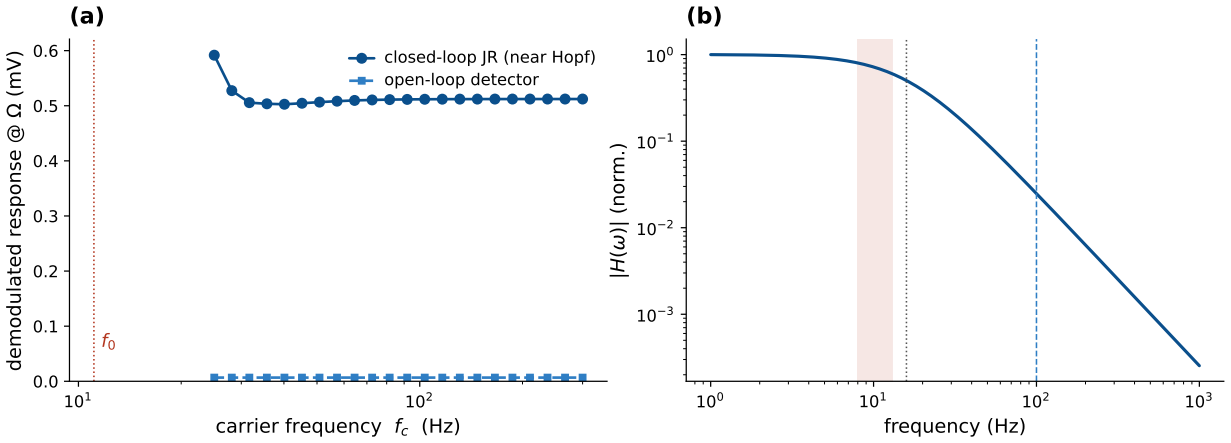


Figure 5: The demodulated response is carrier-independent at fixed polarization (Jansen–Rit). *(a)* The demodulated response is flat across carrier frequency $f_c = 25$–300 Hz, both for the open-loop detector (exactly, by Eq. (10)) and for the closed-loop JR near the Hopf (which adds $\sim 60\times$ resonant gain); deviations appear only as f_c approaches the natural band. *(b)* The second-order-synapse low-pass $|H_s(\omega)|/|H_s(0)|$ passes the envelope (~ 0.7) and suppresses the 100 Hz carrier ($\sim 2.5\%$). Dotted line: synaptic corner $a/2\pi \approx 16$ Hz; shaded band: envelope (8–13 Hz); dashed line: 100 Hz carrier.

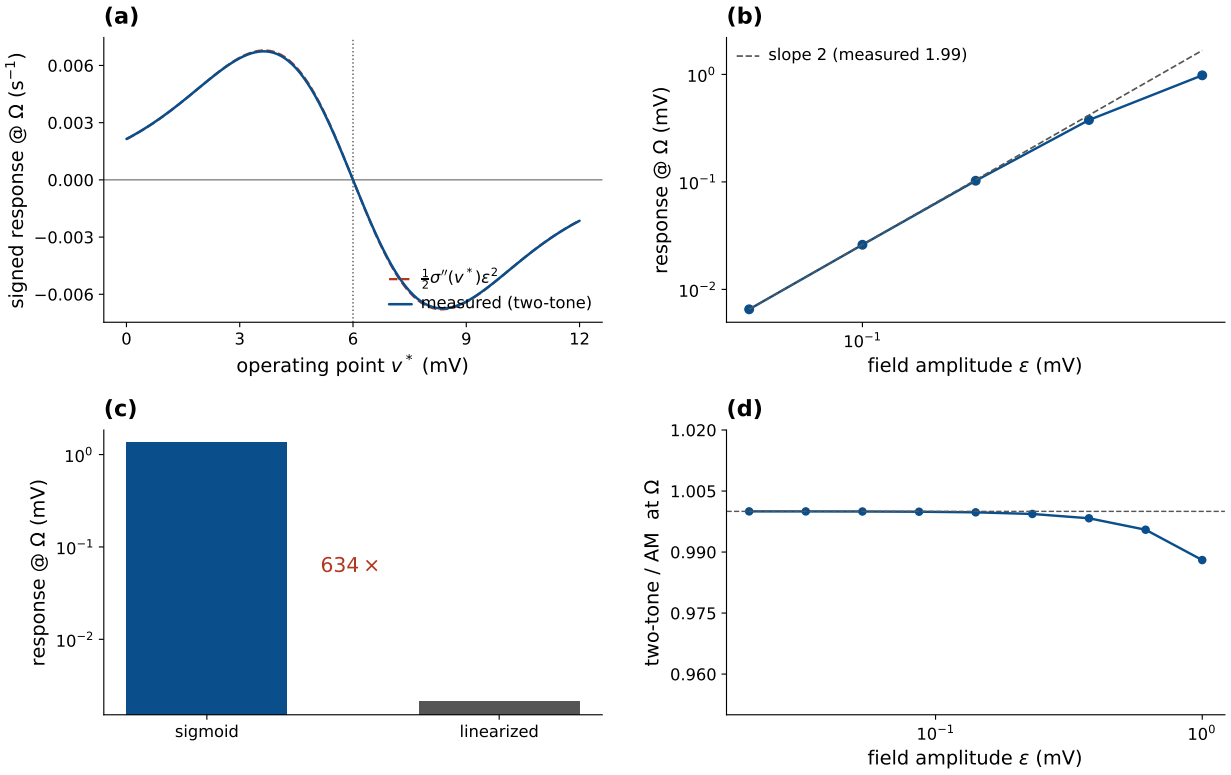


Figure 6: The curvature law, the square law, and the surrogate control (Jansen–Rit, genuine two tones). *(a)* The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2$ across the operating range (maximum deviation 0.9%), vanishing at the inflection v_0 and reversing sign across it. The mechanism is sign-agnostic: σ'' sets sign and phase while $|\sigma''|$ sets gain, so the resonance picture is unchanged up to an overall sign in the $\sigma'' > 0$ lobe, which this sweep spans. The sign flip is itself a prediction (§4.9). *(b)* The closed-loop response scales as ε^2, measured slope 1.99. *(c)* Replacing the field-receiving sigmoid by its tangent collapses the response 634×. *(d)* Surrogate control: the two-tone/AM ratio at Ω approaches 1 as $\varepsilon \to 0$ (1.0000 at 0.02 mV, 0.988 at 1 mV), licensing the AM surrogate for the dense sweeps of Fig. 9, which read only that component, and for nothing else.

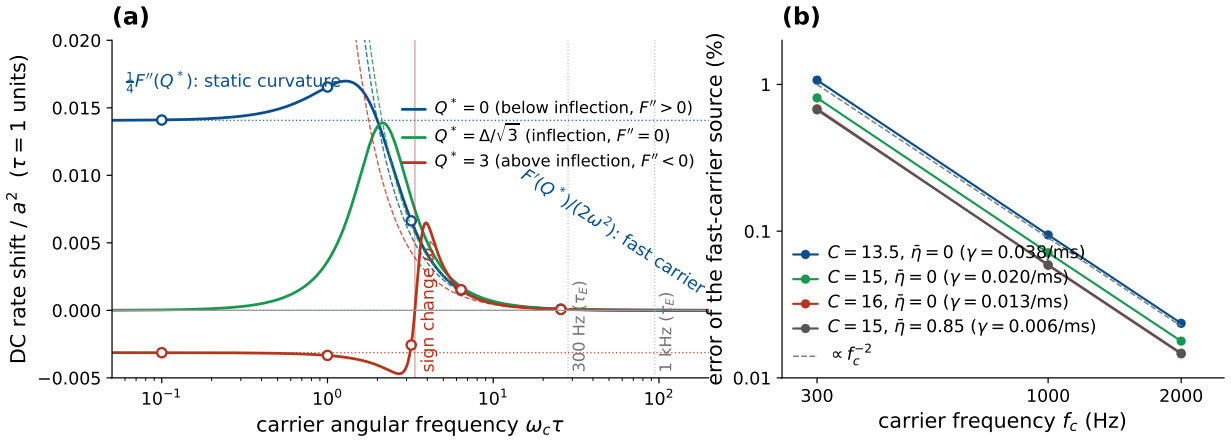


Figure 7: Static curvature at slow carriers, slope at fast ones. *(a)* Second-order DC rate shift of an uncoupled MPR population ($\Delta = 1$, $\tau = 1$) under a current carrier, Eq. (13), for operating points below, at and above the inflection of $F(Q)$. Dotted: the static limit $\frac{1}{4}F''$; dashed: the fast-carrier limit $F'/(2\omega^2)$; circles: RK4 integrations at $a = 0.05$ averaged over whole carrier periods (relative error below 10^{-3}). Above the inflection the sign changes at ω_z. The top axis converts $\omega\tau$ to a carrier frequency for $\tau = \tau_E = 15$ ms. *(b)* PING of Eq. (27): relative error of the fast-carrier difference-frequency source against the exact second-order response at four stable operating points (γ the damping of the gamma mode), two tones centered at 300, 1000 and 2000 Hz.

Two controls test the proposed source of the beat component (Fig. 6b,c): the response scales as ε^2 at small drive (measured log–log slope 1.99 against the predicted 2), and replacing the field-receiving sigmoid by its linearization collapses the response by 634× (1.34 → 0.0021 mV)—the nonlinearity is the detector, and its curvature is the gain. A third panel checks the surrogate at one operating point: for the open-loop detector at $v^* = 8$ mV the two-tone and AM responses at Ω agree to within 10^{-4} at $\varepsilon = 0.02$ mV and to 1.2% at 1 mV (Fig. 6d). On that basis the dense perturbative sweeps below, which read only the Ω component, use the surrogate; the comparison does not test the recurrent response near the Hopf, where the two waveforms' different DC terms could shift the operating state.

4.5 Exact mean-field confirmation: the NMM2 PING gamma generator

The JR column and the LaNMM both use the heuristic firing-rate sigmoid (NMM1). As an independent test we move to the exact next-generation mean field (NMM2, Eq. (4)), where the demodulating nonlinearity is the *derived* quadratic v^2 term rather than a postulated curve. We drive the two-population E–I PING motif of Eq. (27) (Methods), which loses stability through a supercritical Hopf at $\bar{\eta}_H = 1.110$, $f_0 = 54.0$ Hz, with the two-tone field entering the excitatory current, so the carrier is rectified by the exact v_E^2 term.

Driving with a two-tone TI field (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) and locking in the excitatory rate r_E at Δf reproduces the generation and amplification of the beat (Figs. 8, S8). Below the Hopf the response is a forced resonance at the gamma frequency whose height grows as the bifurcation is approached, following the $1/\gamma$ gain of Eq. (20); above it the demodulated drive entrains the autonomous gamma. The two-dimensional map traces a ridge at $\Delta f \approx f_0$ that intensifies through the Hopf. Compared with the JR sweep shown here, the NMM2 resonant frequency shifts appreciably with the operating point, producing a diagonal ridge [20]. That the beat is generated and resonantly amplified with the nonlinearity *derived* from QIF spiking dynamics, rather than assumed, shows the receiver behavior is not an artifact of the heuristic sigmoid. The detector coefficient, however, is not the sigmoid's: at this 300 Hz carrier the population is on the fast side of its crossover, and the generated source is the v_E^2 mixing of the carrier-frequency voltage excursion, operating-point independent to about 1% (§2.3, Fig. 7).

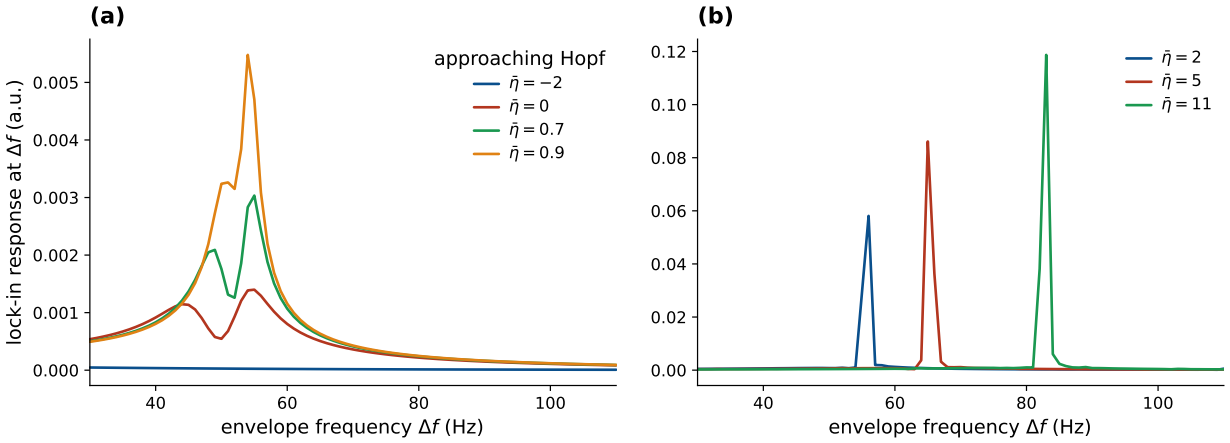


Figure 8: NMM2 PING gamma resonance (exact mean field, genuine two tones). Lock-in response of the excitatory rate r_E at the envelope frequency Δf under a two-tone TI field (carrier 300 Hz). *(a)* (stable focus, $\bar{\eta} < \bar{\eta}_H = 1.110$) forced resonance at the gamma frequency ($f_0 \approx 54$ Hz), the peak growing $0.0014 \to 0.0030 \to 0.0055$ as the Hopf is approached ($1/\gamma$ gain). *(b)* (limit cycle, $\bar{\eta} > \bar{\eta}_H$) entrainment of the autonomous gamma, the peak shifting upward with $\bar{\eta}$ ($56 \to 65 \to 83$ Hz). The input carries no power at Δf; the response is synthesized by the exact quadratic v^2 nonlinearity, with no sigmoid anywhere in the model. Matched AM and two-tone drives agree at Ω to 0.07% well below the Hopf and to 4% at $\bar{\eta} = 0.9$, consistent with the JR surrogate control (Fig. 6d).

4.6 What the mesoscale adds: coupling sets the gain at fixed detection

The resonance sweeps above drove the column toward the Hopf with the external input p, but p also shifts the operating point, entangling detection (σ'') with amplification. To isolate amplification, we vary the JR global connectivity C (coupling strength J) and solve analytically for the drive $p(C)$ that *pins* the operating point v^*, so the detection gain $\sigma''(v^*)$ and the open-loop detector amplitude $A_{\text{open}} = \frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ are held constant. As $C \to C^*$ (the Hopf, $C^* \approx 137$) the damping $\gamma \to 0$, the demodulated response grows $\sim 30\times$ and the envelope resonance sharpens (Fig. 9a,b), a network effect since detection is frozen and only γ changes. The recovered gain $A_\Omega / A_{\text{open}}$ tracks $1/\gamma$ before saturating near criticality (Fig. 9c); driving the same column to the Hopf through the input p rather than the coupling C yields the same gain–γ relation, so the amplification depends only on the distance to criticality, not on which parameter delivers it.

All these sweeps stay on the stable-focus side ($\gamma > 0$), approaching the supercritical Hopf from below: the $1/\gamma$ growth is the forced-resonance susceptibility of a damped network, not entrainment of an autonomous cycle. On the other side of the bifurcation the column oscillates on its own, and the demodulated drive instead phase-locks that limit cycle within an Arnold tongue (Fig. S4; also the entrainment side of Fig. 4 and the LaNMM and NMM2 maps): 1:1 frequency capture over a band that widens with drive, quantified by the locking range rather than a diverging $1/\gamma$ gain, the two coinciding only at onset. Because the drive is weak, on the autonomous side it biases phase (timing) more than amplitude, consistent with the timing-not-rate signature (§4.7).

The next-generation theory gives the same result without pinning. In the NMM2 PING the coupling C is the *literal* mean inter-neuron (QIF) synaptic weight J, and at this carrier the generated source is operating-point independent to about 1% (§2.3), so detection needs no pinning. Sweeping J at fixed background drive $\bar{\eta}$ toward the gamma Hopf J^*, the demodulated peak of r_E grows as $1/\gamma$ and saturates at criticality (Fig. S10): the heuristic JR result, reproduced with the coupling *and* the rectifier both derived rather than assumed, and a direct counterpart of the coupling-controlled weak-field sensitivity of QIF networks [20]. The two columns exhibit the same gain scaling in their two native bands, alpha (JR) and gamma (NMM2): the single-cell nonlinearity detects the envelope (coarse-grained into σ'', or the exact v^2), while the gain is a recurrent amplification set by proximity to the Hopf and adjustable through the coupling.

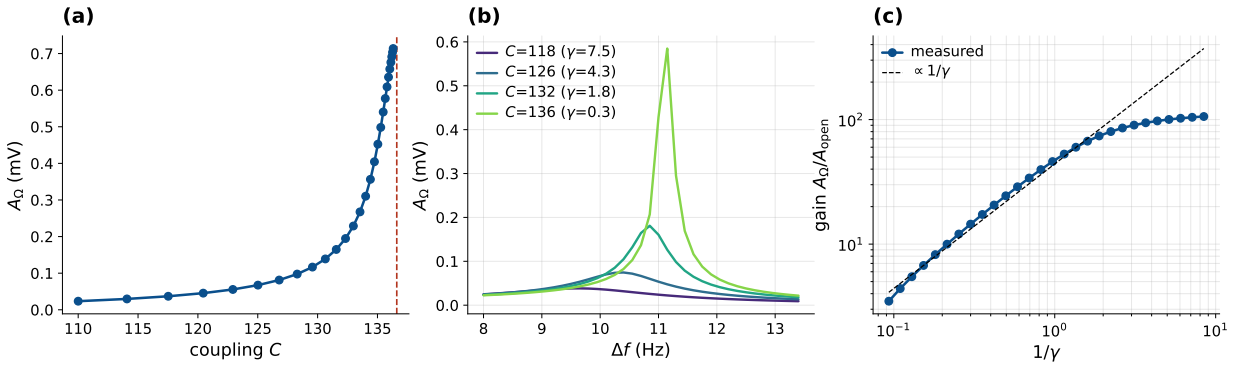


Figure 9: Coupling controls the amplification at fixed detection (Jansen–Rit). The operating point v^* is pinned by co-solving $p(C)$, so $\sigma''(v^*) = -0.151$ and A_{open} are held constant throughout. *(a)* The demodulated A_Ω grows as the coupling $C\,(= J)$ approaches the Hopf C^* from the stable-focus side ($\gamma > 0$). *(b)* The envelope resonance sharpens and grows as $C \to C^*$. *(c)* The network gain A_Ω/A_{open} follows $1/\gamma$ (dashed) and saturates near the bifurcation.

4.7 The resonance amplifies the envelope-locked modulation more than the mean rate

The demodulator feeds the network a constant term and a line at the beat Ω (Eq. (9)), and the transfer $G(\Omega)$ of Eq. (20) treats them differently: $G(\omega_0) \propto 1/(2\gamma\omega_0)$ diverges as the Hopf is approached, while $G(0) \propto 1/\omega_0^2$ stays finite. Pinning v^* to hold the local detector curvature fixed, as in Fig. 9, and driving the column toward the Hopf through the coupling, we read both quantities from the same observable, the pyramidal rate $r_P = \sigma(v + s)$ with the field in its argument. The beat-locked modulation of r_P at $\Delta f = f_0$ grows 13.5× (26 to 346 mHz), tracking $1/\gamma$ before saturating near criticality; the shift of its mean grows too, 5.8× (−1.3 to −7.7 mHz), through rectification of the growing oscillation, but stays below 0.2% of the 3.9 Hz baseline (Fig. S11). Modulation exceeds mean shift by 19× far from the Hopf and by 45–66× near it. The network amplifies the envelope-locked modulation far more than the mean rate, so a small mean-rate change is the expected signature even where TI is acting, the mesoscopic counterpart of the observation that TI alters spike timing more than firing rate in the primate brain [8]. The mean shift is small but nonzero, so the comparison must use commensurate measures, as here.

4.8 Microscale confirmation: TI realigns spike timing in the QIF network

The mesoscale prediction has a direct microscale test, since NMM2 is the *exact* mean field of an all-to-all network of quadratic integrate-and-fire (QIF) neurons with Lorentzian-distributed excitabilities [18, 15, 20]. Integrating that spiking network ($N = 8000$ neurons per population, parameters identical to the PING mean field; Methods) shows TI bunching spike *times* into the beat cycle with a smaller change in the mean rate, on both sides of the gamma Hopf (Fig. 10). Below the Hopf (forced) the field-free network fires asynchronously at a flat rate; with TI on, spikes condense into envelope-locked bands (Fig. 10a,b). Above the Hopf (oscillatory) the free-running gamma is *gated* by the TI beat, a forced cross-frequency modulation (phase–amplitude coupling, PAC), not frequency entrainment, since the ≈11 Hz detuning lies outside the 1:1 tongue: its bursts group into the $\Delta f = 42$ Hz envelope while the intrinsic ∼53 Hz gamma persists (Fig. 10c,d).

Because the two rhythms coexist in a nonlinear medium, the QIF nonlinearity mixes them into intermodulation sidebands in the population rate—most prominently the direct Δf line and the sum component near 95 Hz (53+42), with the difference tone near 11 Hz (the detuning itself); this is the spiking-network fingerprint of the same nonlinear frequency mixing that operates at the macroscale, and it is the sideband signature of *gating* rather than the frequency pulling of entrainment. The spiking network reproduces the exact NMM2 mean field's gamma spectrum (Fig. S12a).

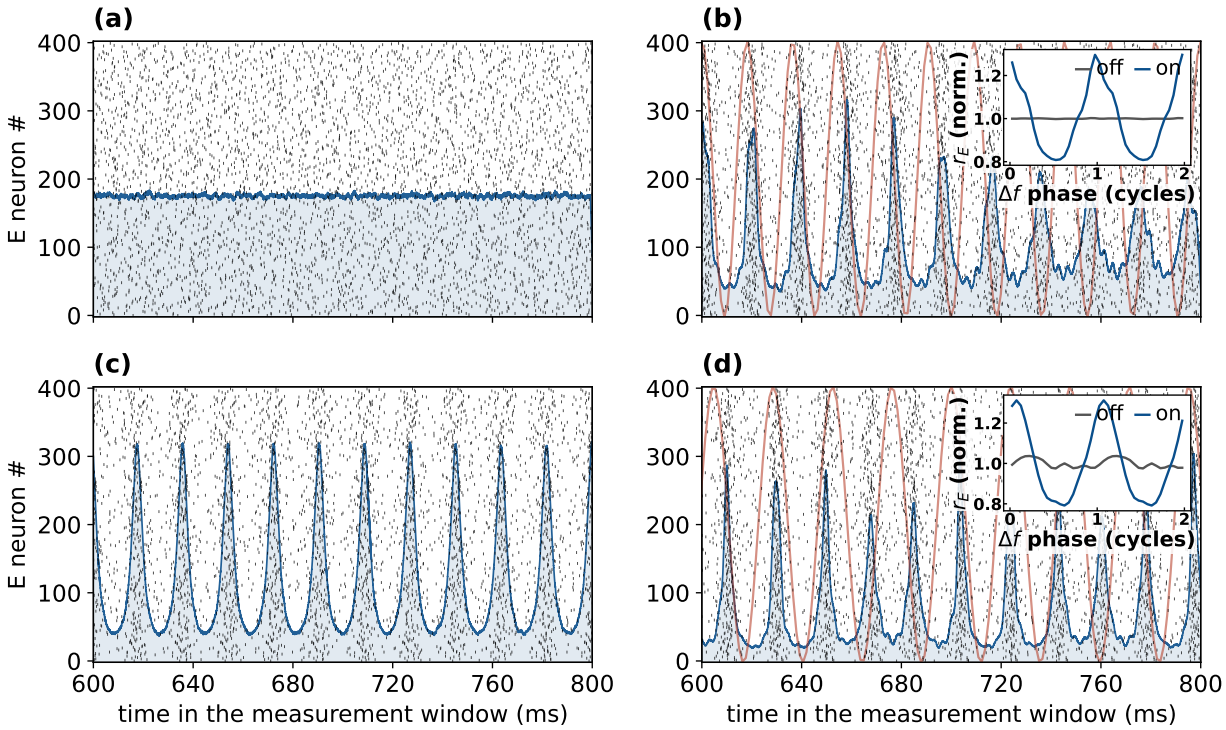


Figure 10: QIF spiking network: TI bunches spikes into the beat cycle with a smaller change in mean rate. $N = 8000$ excitatory and inhibitory quadratic integrate-and-fire neurons whose exact mean field is the NMM2 PING circuit (Eq. (27)). The circuit's intrinsic rhythm is gamma (∼53 Hz), so the beat is set in the gamma range (carrier $f_c = 300$ Hz, genuine two tones). Black ticks are E spikes (400 of 8000), blue the population rate, red the TI envelope; the bifurcation parameter is the common drive $\bar{\eta}$ (gamma Hopf at ≈1.1). *(a,b)* Forced ($\bar{\eta} = 0.85$), TI off then on at $\Delta f = 55$ Hz: asynchronous firing becomes envelope-locked bunching. *(c,d)* Oscillatory ($\bar{\eta} = 1.5$), TI off then on with the beat detuned to 42 Hz: the free-running gamma persists but gains a Δf-locked component. Because the ≈11 Hz detuning places the beat outside the 1:1 tongue, this is forced cross-frequency *gating*, not frequency capture (contrast Fig. S4); the detuning is what certifies the locked component as a driven effect. The last 200 ms of the 800 ms measurement window are shown. *Insets* in *(b,d)*: the population rate folded on the beat phase over the whole window (44 and 34 cycles), each normalized to its own mean (off gray, on blue); with that many cycles the near-commensurate 53 Hz gamma averages out of the 42 Hz fold in *(d)*. Both are flat with TI off.

Across both regimes the mean rate rises by 8–10%, while the Δf modulation depth of the population rate, $A_\Omega/\langle r_E \rangle$, rises from below 0.003 to 0.29 (forced) and 0.23 (gated) (Fig. S12c), measured over 44 and 34 whole beat cycles after 800 ms of settling (Methods). Timing modulation therefore exceeds the relative mean-rate change by a factor of about three at the population level. Per neuron the locking is weak: the pairwise phase consistency of single-neuron spike phases at Δf rises from its field-free control (median −0.02) to a median of 0.003–0.005, and the pooled vector strength from below 0.02 to 0.12–0.13. The population's beat-locked component is built from many weakly biased cells, not from strongly locked ones. This is the spiking-level counterpart of §4.7 and a qualitative analog of the in-vivo finding that TI alters spike timing more than firing rate [8].

4.9 Falsifiable predictions and tests

The mechanism makes four predictions, each with an experimental handle.

1. **Carrier dependence separates access from mixing.** In the static limit and in the fast-carrier QIF asymptote, the leading beat is independent of f_c at fixed local polarization, provided the carrier does not excite another network mode. At fixed applied field, access restores frequency dependence (Eq. (22)); detector dynamics can add it near the crossover. *Test:* measure or estimate local polarization during a carrier sweep and compare the response before and after

correcting for access. The remaining frequency dependence constrains the mixing mechanism. Its cutoff is to be measured, not assigned to a particular compartment in advance.

2. **State- and carrier-dependent response.** The generated beat depends on the operating point through the detector coefficient, and the two modeled detectors disagree on how: the memoryless sigmoid follows $\sigma''(v^*)$ and reverses sign across the inflection, where PEIX (Eq. (24)) peaks; the dynamic mean field at a fast carrier mixes in the voltage dynamics and is read through the rate and network susceptibility; its DC rectification follows F', without the same curvature sign rule (§2.3). *Test:* vary tonic drive and carrier frequency while measuring both amplitude and phase of the beat response. Compare these with the predicted mixing coefficient after accounting for changes in network susceptibility; a response-amplitude change alone does not isolate the detector.

3. **Transduction class.** A smooth memoryless response with nonzero curvature transduces A^2 at weak input, whereas an ideal envelope follower transduces A (§2.4), and the two are separable at a single matched target. The former gives $Y_\Omega \propto \varepsilon^2$ with a second-harmonic ratio vanishing as $\varepsilon \to 0$; the latter gives $Y_\Omega \propto \varepsilon$ with the ratio pinned at $1/5$. *Test:* sweep field amplitude off resonance (so that both lines sit on the flat part of G) and read the log–log slope together with $Y_{2\Omega}/Y_\Omega$; on resonance the network suppresses the second harmonic by $\sim 3\mathcal{Q}$ and the test must be corrected by an independently measured $G(2\Omega)/G(\Omega)$.

4. **Frequency selectivity and state-dependence.** The response is largest when Δf matches a region's resonant rhythm and is amplified as $1/\gamma$ near a bifurcation. Changes in operating state can therefore alter both the generated drive and the recurrent gain [13]. *Test:* sweep Δf through a region's intrinsic band across brain states and track the resonant peak and its width.

Table 3 compares the predictions and related model consequences with existing evidence. Two are qualitatively consistent with published observations: TI alters spike timing more than firing rate in vivo [8] (our $G(\omega_0) \gg G(0)$ asymmetry, §4.7–4.8), and most pyramidal TI responses vanish when synaptic transmission is blocked [9] (a network read-out, which tests network involvement rather than excluding a surviving cellular detector). A quantitative reanalysis of these datasets against the model is a natural next step, and §5 specifies the amplitude-exponent fit it would require.

5 Discussion

Weak-field TI admits an effective description in which cellular nonlinearities generate a quadratic mixing response and recurrent dynamics amplify it. For a smooth response with a nonzero second-order term, microscopic contributions can be combined in the mixing coefficient; identifying each channel is unnecessary for this leading-order account. QIF provides a derived realization of that structure. This connects cellular mixing [2, 4] with the network dependence seen in slices and simulations [9, 10, 11].

Small amplitude and fast relaxation justify different reductions. Weak input permits a second-order expansion of the dynamic response. Fast membrane relaxation gives the NMM2 slow-synapse limit, a static rate curve with dynamic synapses; if relaxation is also fast relative to the carrier, its curvature gives the leading beat. Otherwise the mixing coefficient retains frequency dependence. The curve's slope converts a slow input into a rate change and sets local recurrent gain. The generic quadratic structure survives this distinction, while its magnitude, phase, and frequency dependence remain model-specific.

Fast access has several possible anatomical and electrical origins. Nodes and the AIS contain voltage-gated channels capable of nonlinear rectification [2, 35]; direct field–protein structural effects are reviewed mainly at higher fields, with fields below 0.1 kV/m outside that review's scope [36]. Presynaptic terminals can polarize more strongly than somata [37, 38, 39], but small size alone does not imply fast charging: for an isopotential membrane, the product of specific resistance and

Prediction (model basis)	**Existing evidence**	**Proposed test**
Transduction class: $Y_\Omega \propto \varepsilon^2$ and vanishing $Y_{2\Omega}/Y_\Omega$ (square law) vs. $\propto \varepsilon$ and 1/5 (envelope following)	— (untested; existing comparisons match extracellular field, not ε)	Amplitude sweep off resonance; transfer-corrected harmonic ratio
Timing-dominated response: the envelope-locked modulation of the rate grows toward the Hopf faster than its mean ($G(\omega_0) \gg G(0)$)	*Qualitatively consistent*: TI alters spike timing more than firing rate, primate [8]	Lock-in of spike timing at Δf vs. the mean-rate change, in the same effect measure
Network read-out (population nonlinearity + synaptic recurrence)	*Qualitatively consistent*: most pyramidal TI responses vanish under synaptic block [9]	Synaptic block should reduce recurrent amplification; cellular mixing may persist
Carrier independence at fixed polarization (no ω_c in A_Ω)	TI thresholds inherit the carrier's f_c-dependence [5, 6]; bare kHz weakly effective [34]	Fix in-zone polarization, sweep $f_c \to$ flat envelope response in the stated limits; fit access and residual detector dependence
State- and carrier-dependent response (the detector coefficient is model-dependent)	Cell-type-specific TI responses (PV vs. pyramidal) [9]	Shift v^* by state at fixed beat; sweep f_c at fixed beat
Frequency selectivity, sharpened $\sim 1/\gamma$ near criticality	Quantitative gain–bandwidth relation not tested here in biological data	Sweep Δf through a region's band across brain states

Table 3: The mechanism's predictions confronted with existing in-vivo / in-vitro evidence. Entries cite documented qualitative findings; quantitative reanalysis is left to future work.

capacitance is independent of area. Conductance, geometry, and the mode of stimulation determine the response.

Cohen et al. [40] established a conducting periaxonal space approximately 12 nm wide, partly sealed at the paranodes, and modeled the distinct potentials across myelin and the axonal membrane. Their methods report a minimum fitted axonal decay time near 1 ms; this is not a measured TI access time. Luff et al.'s pulse-width-modulated TI (PWM-TI) model likewise includes a 1 ms passive response within a 1–100 ms parameter sweep [41]. Separately, Semenov et al. [7] measured field-induced membrane charging near 0.7 μs in isolated neurons under strong, microsecond field pulses. This is a different response mode from intracellular current charging or periaxonal conduction. Together these studies motivate examining a shorter effective access time, while leaving its value and field coupling preparation-dependent. A 10–100-fold reduction in a QIF membrane time is therefore a sensitivity hypothesis. Its population-level consequence can be studied through the existing slow-synapse limit, with each physical filter included once; a new periaxonal circuit is not needed for that limiting argument.

A spatial TI dose map depends on this physiology. The usual envelope-modulation measure $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$ assumes that neural response is determined by the envelope excursion. In the local square-law regime, the beat instead depends on $\varepsilon_1\varepsilon_2$, and the carrier average also contains a DC term proportional to $\varepsilon_1^2 + \varepsilon_2^2$. Orientation, access, operating point, and network susceptibility further shape the response. For a smooth memoryless transfer, carrier averaging is even in the envelope amplitude, giving quadratic transduction when the second-order coefficient is nonzero. Dynamic nonlinearities can also mix at second order, with frequency-dependent coefficients. A sharp rectifier can instead approximate envelope following over a finite amplitude range; smoothing or heterogeneity can restore quadratic behavior at smaller inputs. The amplitude and harmonic tests in §2.4 distinguish these regimes.

Stimulation intensity can also change the mechanism. At high intensities, TI and unmodulated kHz fields can have similar excitation thresholds, with modulation patterning firing rather than lowering the threshold [5, 7]; some peripheral responses do not require envelope extraction [6]. The weak-input calculation here concerns modulation of ongoing activity through mixing and amplification [8, 42]. Direct carrier-evoked spiking and conduction block are outside its scope. Finite-amplitude model examples illustrate modulation and saturation beyond the expansion; they do not establish weak-field physiological thresholds.

The recurrent amplifier is shared with direct low-frequency stimulation. In our coupling sweep, holding the JR operating point fixed holds the local curvature fixed while the network gain changes. A direct in-band drive shows the same $1/\gamma$ growth and resonance sharpening (Fig. S3), consistent with the known resonant susceptibility of recurrent excitatory–inhibitory networks [43, 20]. Near the Hopf, this is temporal critical slowing in a mean-field model, not evidence of a diverging spatial correlation length. Related models already reproduce tACS entrainment and AM-driven phase locking [44, 45, 46]. The separation here is between the drive generated by mixing and the response to that drive. When TI and tACS are matched for effective in-band input, observable, and operating state, they share the amplifier. Resonance alone therefore gives TI no efficiency advantage.

Absolute response estimates remain sensitive to access and transduction. Measured static field-to-voltage couplings are 0.12–0.21 mV per V/m in hippocampus [47, 48] and reach about 0.5 mV per V/m in cortical L5/6 pyramidal cells [49]; human cortical fields near 0.8 V/m have been reported at 2 mA [50, 51]. These static couplings do not fix the coupling at kHz. For equal TI carrier polarizations ε, the JR beat drive is $\sigma''\varepsilon^2/2$, whereas a direct drive of the same peak polarization gives $2\sigma'\varepsilon$. Their leading amplitude ratio is $\eta_{\mathrm{det}} \simeq |\sigma''|\varepsilon/(4\sigma')$: the relative efficiency decreases as the input is weakened in this perturbative regime. A larger recurrent gain amplifies both drives.

For scale, a reverberation time $\tau_{\mathrm{rev}} = 1/\gamma$ gives a near-Hopf quality factor $\mathcal{Q} \simeq \pi f_0 \tau_{\mathrm{rev}}$ and power bandwidth approximately $1/(\pi\tau_{\mathrm{rev}})$. Population relaxation over a few hundred milliseconds [52] would give a quality factor of order ten at 10 Hz and a bandwidth of order 1 Hz. This is an order-of-magnitude comparison: the measured relaxation is not specific to the alpha mode. Narrow amplification also makes detuning consequential.

Primate measurements illustrate both the relevance and the limits of an efficiency comparison. Vieira et al. [8] used modulation depths of 0.5–0.62 V/m and reported TI effects at 0.34 times those of conventional tACS (95% confidence interval 0.04–0.65). Their within-electrode AM comparison compensated for frequency-dependent current shunting and matched extracellular modulation depth near the recorded cells. A favorable fast-access estimate with $\varepsilon \approx 0.1$ mV gives only about 0.8% for the JR detector, below the reported ratio. This discrepancy is not a calibrated rejection of that detector: the comparison involved twenty neurons, with ten responding to at least one modality and seven to both; phase-locking value (PLV) depends on baseline locking and network state; and the matched extracellular fields do not ensure matched polarization at the cellular mixer. An unspecified nonlinear relation between generated drive and PLV also prevents identification of the input-amplitude exponent. The data establish a weaker AM effect than the matched-depth low-frequency drive, but do not isolate the transduction law.

For comparison, the matched-peak efficiency over the logistic family has a supremum of approximately 0.348 (App. D). Approaching it at weak fields requires a transfer curve narrowed to microvolt scales; the limiting detector is step-like and has a second-harmonic ratio near 0.15, distinct from both weak square-law detection and linear envelope following. This bound applies to that memoryless family and matching convention, not to the dynamic QIF detector or to all cellular mechanisms. Its proximity to an experimental ratio is not a fit.

Quantitative physiological predictions require calibration of access and mixing. Their values are not fixed by the quadratic scaling law. In comparisons of stimulus-evoked firing, Hodgkin–Huxley

and Frankenhaeuser–Huxley models reproduce responses missed by the tested leaky, exponential, and adaptive-exponential integrate-and-fire models [53]. Such model dependence concerns the specific response, without invalidating the weak-input expansion. Noise, heterogeneity beyond the assumed excitability distribution, and interactions between columns can broaden or shift the deterministic resonances. Cascading ordinary synaptic filters cannot solve carrier access, because they attenuate the carrier before a subsequent nonlinearity receives it. Human results are also mixed, including reported peripheral effects without central effects [54] and null occipito-parietal alpha-power modulation [55]. Testing the proposed mechanism requires measuring access, generated beat, and network tuning in the same preparation.

6 Conclusion

In the weak-field regime, smooth cellular nonlinearities with a nonzero second-order response can be represented by effective quadratic mixing. The population models provide realizations of this description: mixing generates the difference-frequency component, and recurrent dynamics amplify it near a network resonance. A large network response does not, by itself, imply efficient detection or identify the microscopic mixer.

The static and dynamic population descriptions are connected by their time scales. Fast membrane relaxation gives the NMM2 slow-synapse limit: a nonlinear static rate map coupled through dynamic synapses. Its curvature describes carrier mixing when relaxation is also fast relative to the carrier. Outside that limit, the dynamic response must be used. Fast cellular access and slow recurrent resonance can therefore coexist, with each physical filtering process represented once.

In the weak square-law regime, beat generation depends on the product of the effective carrier inputs, not only on the envelope excursion. TI and a direct low-frequency drive share the downstream amplifier when input, observable, and state are matched, so resonance alone gives TI no efficiency advantage. The anatomical access pathway and absolute response at human field strengths remain open. Measuring access, mixing, and circuit tuning separately tests the proposed mechanism without attributing all three to a single transfer curve.

Code and data availability

All simulation and figure-generation code, the committed datasets, and a one-command reproduce script (`run_all.py`) are openly available at the paper's GitHub repository, archived at Zenodo (doi:10.5281/zenodo.21009618). The repository `README` documents the per-figure script mapping (a figure→script table) and the dependency order, so every figure in this note can be regenerated from the committed inputs. The code is released under CC-BY-4.0.

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Competing interests

GR is Chief Scientific and Technology Officer of Neuroelectrics, which develops and commercializes transcranial stimulation devices. BM, AJ and FC are affiliated with Neuroelectrics.

Author contributions

Using the CRediT taxonomy: **G. Ruffini**: conceptualization, methodology, formal analysis, software, investigation, visualization, writing (original draft), supervision, funding acquisition. **B. Mercadal**: validation, visualization, writing (review and editing). **A. Just**: software, visualization, writing (review and editing). **R. de Palma Aristides**: software, visualization, writing (review and editing). **F. Castaldo**: conceptualization, writing (review and editing). **N. Grossman**: conceptualization. **S. Canals**: supervision, writing (review and editing). **C. Mirasso**: supervision, writing (review and editing).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic Claude and OpenAI ChatGPT to streamline the narrative and to assist with code and figure generation from data the authors produced. The authors reviewed and edited all resulting content and take full responsibility for the content of the published article.

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